Controllability of $\mathcal{N}$-Duplication Corona Product Networks With Laplacian Dynamics

Bo Liu, Xuan Li, Junjie Huang, and Housheng Su

***Abstract*— This article studies the controllability of a new composite network generated by two smaller scale factor networks via the Corona product with Laplacian dynamics. First, the eigenvalues and corresponding eigenvectors of a new composite network—the $\mathcal{N}$-duplication Corona product network—are derived by some properties of its factor networks. Second, a necessary and sufficient algebra-based criterion for the controllability of such network is established based on the Popov-Belevitch-Hautus (PBH) test. Furthermore, the weights on edges between the different factor networks are considered. Finally, several examples are presented to demonstrate the effectiveness of our results applied to the unmanned aerial vehicle (UAV) formation.**

***Index Terms*— Controllability, Corona product networks, Laplacian dynamics, multiagent networks (MANs).**

Nomenclature

$\circ$	(1-duplication) Corona product.
$\divideontimes$	2-duplication Corona product.
$\underline{\divideontimes}$	$\mathcal{N}$-duplication Corona product.
$\otimes$	Kronecker product.
$\mathcal{R}$	Real number field.
$\mathcal{C}$	Complex number field.
e_i	Unit vector.
I_n	n-dimensional identity matrix.
1_n	Vector with all elements being 1.

I. Introduction

A CONTROLLABLE system is a prerequisite for further studying the performance index, optimization control, and other problems of the system. The controllability is one of the ultimate goals of studying the models, structures, and dynamics of multiagent networks (MANs), which has attracted great attention and has been applied in engineering areas, such as unmanned aerial vehicle (UAV) formation control, power system control, and urban traffic networks [1], [2], [3], [4], [5], [6], [8], [9], [10].

With the increasing size of MANs, the network topology becomes more complex, which is increasingly important for the influence on the overall behavior of the systems [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. The coordinated control of MANs can be investigated from the existing interactions between multiple subnetworks, which involves the properties preserved under the composition of networks. An example of this is to establish the controllable or/and observable conditions for general series–parallel and feedback composite systems in the light of the controllability and observability of their subsystems [26], which provides a compositional perspective for description, analysis, and computation of systems. Ding et al. [27] proposed target minimum-cost maximum flow (TMM) to ensure the target nodes in multiplex networks to be controlled with the minimum number of inputs as well as the maximum (minimum) number of covered nodes. A more accurate method for predicting network controllability robustness was developed using machine learning in [28]. In [29], an equivalent condition for the observability of Boolean control networks (BCNs) with time-variant delays in states was presented by dividing the system into a finite number of subsystems without time-delays.

Graph product, a multiplication operation, is a very effective method for generating a large-scale network from two or more simpler and lower dimensional factor ones and preserving some properties of its factor ones [30], [31], [32], which can offer compact descriptions and mathematical representations of systems. Some interesting graph products, such as direct product and strong product [31], Cartesian product [33], [34], [35], Kronecker product [36], and Corona product [37], can be used explicitly and visually to calculate or estimate the eigenvalues of the Laplacian or adjacency matrices, which offers a useful framework for modeling and analysis of MANs and gives further insight into the influence of network topology on the controllability of MANs.

Until now, due to the complexity of the graph product networks and the difficulty in accurate mathematical description, the study on the controllability of graph product networks is still in its infancy. Chapman et al. [33] studied the controllability and observability of a composite network, which is generated from smaller scale factor networks via Cartesian

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Bo Liu and Xuan Li are with the Key Laboratory for Intelligent Analysis and Security Governance of Ethnic Languages, Ministry of Education, School of Information Engineering, Minzu University of China, Beijing 100081, China (e-mail: boliu2020@muc.edu.cn; 21301972@muc.edu.cn).

Junjie Huang is with the School of Mathematical Sciences, Inner Mongolia University, Hohhot 010021, China (e-mail: huangjunjie@imu.edu.cn).

Housheng Su is with the School of Artificial Intelligence and Automation, Key Laboratory of Image Processing and Intelligent Control of Education Ministry of China, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: houshengsu@gmail.com).

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TABLE I
SOME PROPERTIES OF CORONA PRODUCTS WITH $\mathcal{G}_1$ AND $\mathcal{G}_2$

Type	No. of vertices	No. of edges	Laplacian matrices
$\mathcal{G}_1 \circ \mathcal{G}_2$	$n_1(n_2+1)$	$m_1 + n_1(m_2+n_2)$	$\begin{bmatrix} \mathcal{L}_1 + n_2 I_{n_1} & -I_{n_1} \otimes 1_{n_2}^T \\ -I_{n_1} \otimes 1_{n_2} & I_{n_1} \otimes (\mathcal{L}_2 + I_{n_2}) \end{bmatrix}$
$\mathcal{G}_1 \divideontimes \mathcal{G}_2$	$n_1(n_2+2)$	$2m_1 + n_1(m_2+2n_2)$	$\begin{bmatrix} I_2 \otimes (\mathcal{L}_1 + n_2 I_{n_1}) & -1_2 \otimes (I_{n_1} \otimes 1_{n_2}^T) \\ -1_2^T \otimes (I_{n_1} \otimes 1_{n_2}) & I_{n_1} \otimes (\mathcal{L}_2 + 2I_{n_2}) \end{bmatrix}$
$\mathcal{G}_1 \underline{\divideontimes} \mathcal{G}_2$	$n_1(n_2+\mathcal{N})$	$\mathcal{N}m_1 + n_1(m_2+\mathcal{N}n_2)$	$\begin{bmatrix} I_{\mathcal{N}} \otimes (\mathcal{L}_1 + n_2\omega I_{n_1}) & -1_{\mathcal{N}} \otimes (\omega I_{n_1} \otimes 1_{n_2}^T) \\ -1_{\mathcal{N}}^T \otimes (\omega I_{n_1} \otimes 1_{n_2}) & I_{n_1} \otimes (\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}) \end{bmatrix}$

products and obtained a necessary and sufficient controllability condition with diagonalizable topology matrices. She et al. [34] considered the energy-related controllability of Cartesian product networks. Compared with the classical controllability, referring to the performance that the state variables of a networked system can be controlled by external inputs, the energy-related controllability is to measure the energy needed in network control from an energy perspective in practical applications, which can be characterized by the Gramian controllability matrix. Structural observability and controllability of Kronecker product networks were considered in [36]. Very recently, Wang et al. [37] investigated the controllability of Corona product networks. The Corona product of two graphs has multiple product ways [38], [39], [40], [41], [42] and a wide range of applications, such as UAV formation, virus replication, and some special chemical and biological structures or communities (e.g., quantum molecules, a small polypeptide, and protein–protein interactions) [43], [44], [45]. Khalid et al. [44] analyzed degree-based topological indices for some chemically important molecular graphs via Corona products. However, how to reasonably build a composite network model generated by the factor networks through the Corona product? How to characterize the interactions between the factor networks? What kind of dynamic behavior does the evolution of agents present with the factor networks? What is the effect of the factor networks on the controllability? What is the controllability relationship between factor networks and the composite network? What theoretical methods and analytical tools are used to analyze the controllability? These problems are new challenges and difficulties in studies on the controllability of new composite MANs generated by two factor networks through graph products.

This article establishes the controllability of a composite MAN via Corona products with Laplacian dynamics. The main technical contributions of this work are fourfold. First, a new composite network—the $\mathcal{N}$-duplication Corona product network—is constructed. In most of the existing literature, the first factor network is copied only once and the second factor network is replicated in terms of the number of nodes in the first one. However, in our work, the first factor network is copied $\mathcal{N}$ times, which is more complicated to create a new composite network and analyze the controllability for such MANs. Second, the eigenvalues and corresponding eigenvectors of the $\mathcal{N}$-duplication Corona product network are obtained by some properties of its factor networks. Based on this, a necessary and sufficient algebra-based criterion for the controllability of such network is established. Third, all the edge weights within each factor network and between different factor networks are considered. Finally, our result is applicable to a UAV formation. The research methods and controllability analysis will help us to design and optimize control algorithms for MANs.

The rest of this study is organized as follows. Some related notations and important preliminaries are presented in Section II. Section III states the problem and gives the spectra and eigenvectors of the $\mathcal{N}$-duplication Corona product network. The controllability conditions of the $\mathcal{N}$-duplication Corona product network are established in Section IV. Section V presents several simulation examples. Finally, Section VI draws the conclusion.

II. NOTATIONS AND PRELIMINARIES

A. Notations

Some available notations in this study are given in the Nomenclature.

B. Graphs

A triple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ is an undirected graph with a vertex set $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$, an edge set $\mathcal{E} = \{e_k : e_k = (v_i, v_j), k \in J_m\} \subseteq \mathcal{V} \times \mathcal{V}$, and the weighted adjacency matrix $\mathcal{A} = [a_{ij}] \in R^{n\times n}$. Laplacian matrix $\mathcal{L}$ is defined as $\mathcal{L} \triangleq \mathcal{D} - \mathcal{A}$, where $\mathcal{D}$ is the degree matrix. Since the Laplacian matrix is symmetric and semipositive definite with all $l_{ij} \leq 0$ for all $i \neq j$, $\gamma = 0$ is an eigenvalue of $\mathcal{L}$ with corresponding associated eigenvector $\xi = (1, 1, \ldots, 1)^T = 1_n$, and all the rest eigenvalues of $\mathcal{L}$ are real and positive; therefore, it follows from matrix theory [46] that the eigenvalues of $\mathcal{L}$ can be ordered as $0 = \gamma_1 < \gamma_2 \leq \gamma_3 \cdots \leq \gamma_n$.

C. Corona Products

Let $\mathcal{G}_1$ (n_1 vertices and m_1 edges) and $\mathcal{G}_2$ (n_2 vertices and m_2 edges) be two finite, simple, nonempty, and vertex-disjoint graphs. A composite graph $\mathcal{G} \triangleq \mathcal{G}_1 \circ \mathcal{G}_2$ is said to be the (1-duplication) Corona product of $\mathcal{G}_1$ and $\mathcal{G}_2$ with the vertex and edge sets $\mathcal{V}(\mathcal{G}_1 \circ \mathcal{G}_2) = \mathcal{V}(\mathcal{G}_1) \cup \{n_1 \text{ different copies of } \mathcal{V}(\mathcal{G}_2)\}$ and $\mathcal{E}(\mathcal{G}_1 \circ \mathcal{G}_2) = \mathcal{E}(\mathcal{G}_1) \cup \{n_1 \text{ different copies of } \mathcal{E}(\mathcal{G}_2)\} \cup \{(u_i, v) : u_i \in \mathcal{V}(\mathcal{G}_1), v \in \mathcal{V}(\mathcal{G}_2), i \in \underline{n_1} \triangleq \{1, 2, \ldots, n_1\}\}$, respectively, that is, it is generated by taking one copy of $\mathcal{G}_1$ and n_1 disjoint copies of $\mathcal{G}_2$ and making each vertex in the

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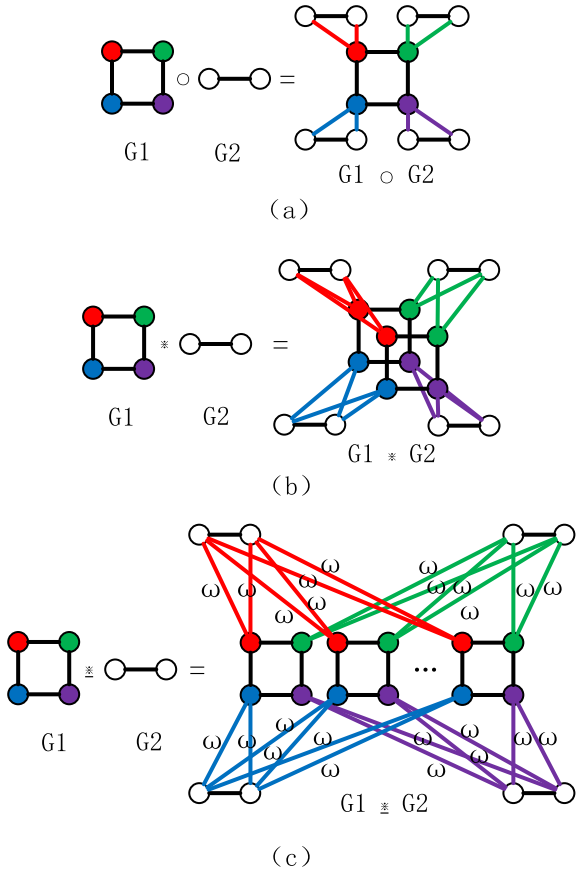


Fig. 1. Corona product graphs of $\mathcal{G}_1$ and $\mathcal{G}_2$, where graphs (a)–(c) indicate 1-duplication, 2-duplication, and $\mathcal{N}$-duplication Corona products of $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively.

ith copy of $\mathcal{G}_2$ adjacent to the ith vertex of $\mathcal{G}_1$ by a new edge for $i \in \underline{n_1}$. Similarly, the composite graph $\mathcal{G}(\triangleq \mathcal{G}_1 \underline{\circledast} \mathcal{G}_2)$ is said to be the $\mathcal{N}$-duplication Corona product of $\mathcal{G}_1$ and $\mathcal{G}_2$, which is obtained by taking $\mathcal{N}$ copies of $\mathcal{G}_1$ and n_1 disjoint copies of $\mathcal{G}_2$ and making each vertex in the ith copy of $\mathcal{G}_2$ adjacent to the ith vertex of each copy of $\mathcal{G}_1$ by a new edge for $i \in \underline{n_1}$. In particular, as $\mathcal{N} = 2$, it is called as 2-duplication Corona product, denoted by $\mathcal{G}_1 \circledast \mathcal{G}_2$. More details can be seen in Table I [38], [39]. An example of the Corona product graph is shown in Fig. 1.

From Table I, it is easy to see that the Laplacian matrix

$$\begin{bmatrix} I_{\mathcal{N}} \otimes \left(\mathcal{L}_1 + n_2 \omega I_{n_1}\right) & -1_{\mathcal{N}} \otimes \left(\omega I_{n_1} \otimes 1_{n_2}^T\right) \\ -1_{\mathcal{N}}^T \otimes \left(\omega I_{n_1} \otimes 1_{n_2}\right) & I_{n_1} \otimes \left(\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}\right) \end{bmatrix}$$

of $\mathcal{N}$-duplication Corona product network is more complex than the Laplacian matrix

$$\begin{bmatrix} \mathcal{L}_1 + n_2 I_{n_1} & -I_{n_1} \otimes 1_{n_2}^T \\ -I_{n_1} \otimes 1_{n_2} & I_{n_1} \otimes \left(\mathcal{L}_2 + I_{n_2}\right) \end{bmatrix}$$

of the simple Corona product network, which will make it more difficult to analyze the controllability of $\mathcal{N}$-duplication Corona product network by the eigenvalues and eigenvectors of its factor networks and reveal the controllability relationship between the whole network and its factors.

D. Some Propositions Available

In this venue, the following propositions are useful to illuminate our results.

Proposition 1: (See [40]) Let $A \in \mathcal{R}^{n_1 \times n_1}$, $B \in \mathcal{R}^{n_1 \times n_2}$, $C \in \mathcal{R}^{n_2 \times n_1}$, and $D \in \mathcal{R}^{n_2 \times n_2}$, and then,

$$\det\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{cases} \det(A)\det\left(D - CA^{-1}B\right), \text{if } A \text{ is invertible} \\ \det(D)\det\left(A - BD^{-1}C\right), \text{if } D \text{ is invertible.} \end{cases}$$

Proposition 2: Let

$$M = \begin{bmatrix} M_1 & M_2 & M_2 & \cdots & M_2 \\ M_2 & M_1 & \ddots & \ddots & M_2 \\ M_2 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & M_2 \\ M_2 & M_2 & \cdots & M_2 & M_1 \end{bmatrix}$$

be an $n \times n$ block matrix. Then, the eigenvalues of M are those of $M_1 + (n-1)M_2$ and $M_1 - M_2$.

Proof: The characteristic polynomial of M is, as shown in the equation at the bottom of the next page, as required. □

Proposition 3: Let $\mathcal{L} \in \mathcal{R}^{n \times n}$ be the Laplacian matrix of $\mathcal{G}$, and then,

$$\Gamma_{\mathcal{L}}(\gamma) = \frac{n}{\gamma}$$

where $\Gamma_{\mathcal{L}}(\gamma) \triangleq 1_n^T(\gamma I_n - \mathcal{L})^{-1}1_n$ is called the $\mathcal{L}$-coronal of $\mathcal{G}$ over the field of rational functions $\mathcal{C}(\gamma)$ (see [41], [42]).

Proof: Since $\mathcal{L}1_n = 0$, then $(\gamma I_n - \mathcal{L})1_n = \gamma 1_n$. Thus, □

$$\Gamma_{\mathcal{L}}(\gamma) = 1_n^T(\gamma I_n - \mathcal{L})^{-1}1_n = \frac{1_n^T 1_n}{\gamma} = \frac{n}{\gamma}.$$

Notice that, viewed as a matrix over $\mathcal{C}(\gamma)$, the characteristic matrix $\gamma I - \mathcal{L}$ has $\det(\gamma I - \mathcal{L}) \neq 0$, so $\gamma I - \mathcal{L}$ is invertible. Notice also that $\gamma = 0$ is the only pole of $\Gamma_{\mathcal{L}}(\gamma) = (n/\gamma)$, which is equal to the minimum eigenvalue of the Laplacian matrix $\mathcal{L}$ of $\mathcal{G}$.

III. Problem Statement

A. Model

Consider the two leader–follower MANs $\mathcal{G}_1$ and $\mathcal{G}_2$, whose dynamics are described by

$$\begin{aligned} \dot{x}_{\mathcal{G}_1} &= -\mathcal{L}_1 x_{\mathcal{G}_1} + \mathcal{B}_1 u_1 \\ \dot{x}_{\mathcal{G}_2} &= -\mathcal{L}_2 x_{\mathcal{G}_2} + \mathcal{B}_2 u_2 \end{aligned} \tag{1}$$

where $x_{\mathcal{G}_i} \in \mathcal{R}^{n_i}$, $u_i \in \mathcal{R}^{d_i}$, $\mathcal{L}_i \in \mathcal{R}^{n_i \times n_i}$, and $\mathcal{B}_i = [e_1, e_2, \ldots, e_{d_i}] \in \mathcal{R}^{n_i \times d_i}$ for $i = 1, 2$ are the aggregated agent state vectors, external control inputs, the Laplacian, and control matrices of the factor networks $\mathcal{G}_1$ and $\mathcal{G}_2$; and d_1 and d_2 are the inputs (the leaders' numbers) of $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively. A compact form of formulating the $\mathcal{N}$-duplication weighted Corona product with factor networks $\mathcal{G}_1$ and $\mathcal{G}_2$ described as (1) is

$$\Sigma: \ \dot{x} = -\mathcal{L}x + \mathcal{B}u \tag{2}$$

where

$$\mathcal{L} = \begin{bmatrix} I_{\mathcal{N}} \otimes \left(\mathcal{L}_1 + n_2 \omega I_{n_1}\right) & -1_{\mathcal{N}} \otimes \left(\omega I_{n_1} \otimes 1_{n_2}^T\right) \\ -1_{\mathcal{N}}^T \otimes \left(\omega I_{n_1} \otimes 1_{n_2}\right) & I_{n_1} \otimes \left(\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}\right) \end{bmatrix},$$

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$$\mathcal{B}=\begin{bmatrix} I_{\mathcal{N}}\otimes\mathcal{B}_1 & 0 \\ 0 & I_{n_1}\otimes\mathcal{B}_2 \end{bmatrix}$$

and $\omega>0$ is the edge weight of each new edge linking $\mathcal{G}_1$ with $\mathcal{G}_2$.

Throughout this work, system $(\mathcal{G},\Sigma)$ denotes the MAN $\mathcal{G}$ consisting of factor networks $\mathcal{G}_1$ and $\mathcal{G}_2$ by $\mathcal{N}$-duplication weighted Corona product and dynamics (2), where $\mathcal{G}=\mathcal{G}_1\underline{\ast}\mathcal{G}_2$. The terms "system $(\mathcal{G},\Sigma)$" and "matrix pair $(\mathcal{L},\mathcal{B})$" [or pair $(-\mathcal{L},\mathcal{B})$] are used interchangeably. Notice that the analysis of this work is based on Proposition 3, which considers the case of Laplacian dynamics satisfying the row sums being zero. For more general networks, such as networks with antagonistic interactions (i.e., the edge weights of agents and ω are arbitrary nonzero and different), the system matrix is not a Laplacian matrix, so these results cannot be directly extended, which will be focused in our future work.

B. Spectra of $\mathcal{N}$-Duplication Corona Product Network

In this section, the complete spectrum features of the $\mathcal{N}$-duplication weighted Corona product network $\mathcal{G}=\mathcal{G}_1\underline{\ast}\mathcal{G}_2$ are revealed by checking some features of its factor networks $\mathcal{G}_1$ and $\mathcal{G}_2$. In fact, in order to calculate the spectra of $\mathcal{G}$, only the coronal of $\mathcal{G}_2$ [i.e., $\Gamma_{\mathcal{L}_2}(\gamma)$] is required, beyond the spectra of $\mathcal{G}_1$ and $\mathcal{G}_2$.

Lemma 1: Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two graphs on n_1 and n_2 vertices, respectively; and all the weights of the newly added edges between $\mathcal{G}_1$ and $\mathcal{G}_2$ are the same and positive, denoted as $\omega>0$. Let $\sigma(\mathcal{G}_1)=\{\lambda_1,\lambda_2,\ldots,\lambda_{n_1}|0=\lambda_1<\lambda_2\le\cdots\le\lambda_{n_1}\}$ and $\sigma(\mathcal{G}_2)=\{\mu_1,\mu_2,\ldots,\mu_{n_2}|0=\mu_1<\mu_2\le\cdots\le\mu_{n_2}\}$ be the spectrum sets of $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively. Then,

$$\Phi_{\mathcal{G}}(\mathcal{L}:\gamma)\triangleq\det(\gamma I-\mathcal{L})$$
$$=\prod_{i=1}^{n_1}\left(\gamma^2-(\mathcal{N}\omega+n_2\omega+\lambda_i)\gamma+\mathcal{N}\omega\lambda_i\right)$$
$$\prod_{j=2}^{n_2}(\gamma-\mathcal{N}\omega-\mu_j)^{n_1}\prod_{i=1}^{n_1}(\gamma-n_2\omega-\lambda_i)^{\mathcal{N}-1}.$$

Moreover, the spectrum of $\mathcal{G}=\mathcal{G}_1\underline{\ast}\mathcal{G}_2$ can be expressed as $\sigma(\mathcal{G})=\Omega_1\cup\Omega_2\cup\Omega_3$, where $\Omega_1=\Omega_{11}\cup\Omega_{12}$ with $\Omega_{11}=\{\delta_{i1}|\delta_{i1}=((\mathcal{N}\omega+n_2\omega+\lambda_i)+((\mathcal{N}\omega+n_2\omega+\lambda_i)^2-4\mathcal{N}\omega\lambda_i)^{1/2})/2, i=1,2,\ldots,n_1\}$ and $\Omega_{12}=\{\delta_{i2}|\delta_{i2}=((\mathcal{N}\omega+n_2\omega+\lambda_i)-((\mathcal{N}\omega+n_2\omega+\lambda_i)^2-4\mathcal{N}\omega\lambda_i)^{1/2})/2, i=1,2,\ldots,n_1\}$; $\Omega_2=\{\mu_j+\mathcal{N}\omega|j=2,3,\ldots,n_2\}$, each eigenvalue in Ω_2 is n_1-repeated; and $\Omega_3=\{\lambda_i+n_2\omega|i=1,2,\ldots,n_1\}$, each eigenvalue in Ω_3 is $(\mathcal{N}-1)$-repeated.

Proof: Combining with properties of Kronecker products and Proposition 1, we can have (as an equality of rational functions) □

$$\begin{aligned}
&\Phi_{\mathcal{G}}(\mathcal{L}:\gamma)\triangleq\det(\gamma I-\mathcal{L})\\
&=\det\begin{bmatrix} I_{\mathcal{N}}\otimes\left((\gamma-n_2\omega)I_{n_1}-\mathcal{L}_1\right) & 1_{\mathcal{N}}\otimes\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\\ 1_{\mathcal{N}}^T\otimes\left(\omega I_{n_1}\otimes 1_{n_2}\right) & I_{n_1}\otimes\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)\end{bmatrix}\\
&=\det\left(I_{n_1}\otimes\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)\right)\det\Big[I_{\mathcal{N}}\otimes\left((\gamma-n_2\omega)I_{n_1}-\mathcal{L}_1\right)\\
&\quad-\left(1_{\mathcal{N}}\otimes\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\right)\left(I_{n_1}\otimes\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)\right)^{-1}\\
&\quad\left(1_{\mathcal{N}}^T\otimes\left(\omega I_{n_1}\otimes 1_{n_2}\right)\right)\Big]\\
&=\det\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)^{n_1}\det\left(I_{n_1}\right)^{n_2}\\
&\quad\det\Big[I_{\mathcal{N}}\otimes\left((\gamma-n_2\omega)I_{n_1}-\mathcal{L}_1\right)-\omega^2\left(\left(1_{\mathcal{N}}\otimes I_{n_1}\right)\otimes 1_{n_2}^T\right)\\
&\quad\left(I_{n_1}\otimes\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)^{-1}\right)\left(\left(1_{\mathcal{N}}^T\otimes I_{n_1}\right)\otimes 1_{n_2}\right)\Big]\\
&=\det\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)^{n_1}\\
&\quad\det\Big[I_{\mathcal{N}}\otimes\left((\gamma-n_2\omega)I_{n_1}-\mathcal{L}_1\right)-\omega^2\left(\left(1_{\mathcal{N}}\otimes I_{n_1}\right)I_{n_1}\otimes\right.\\
&\quad\left.\left(1_{n_2}^T\left((\gamma-\mathcal{N}\omega)I_{n_2}-\mathcal{L}_2\right)^{-1}\right)\left(\left(1_{\mathcal{N}}^T\otimes I_{n_1}\right)\otimes 1_{n_2}\right)\right)\Big]
\end{aligned}$$

$$\begin{aligned}
&\det(\gamma I-M)\\
&=\det\begin{bmatrix} \gamma I-M_1 & -M_2 & -M_2 & \cdots & -M_2\\ -M_2 & \gamma I-M_1 & \ddots & \ddots & -M_2\\ -M_2 & \ddots & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \ddots & -M_2\\ -M_2 & -M_2 & \cdots & -M_2 & \gamma I-M_1\end{bmatrix}\\
&=\det\begin{bmatrix} \gamma I-M_1 & -M_2 & -M_2 & \cdots & -M_2\\ -(\gamma I-M_1+M_2) & \gamma I-M_1+M_2 & 0 & \cdots & 0\\ -(\gamma I-M_1+M_2) & 0 & \ddots & \ddots & \vdots\\ \vdots & \vdots & \ddots & \ddots & 0\\ -(\gamma I-M_1+M_2) & 0 & \cdots & 0 & \gamma I-M_1+M_2\end{bmatrix}\\
&=\det\begin{bmatrix} \gamma I-(M_1+(n-1)M_2) & -M_2 & -M_2 & \cdots & -M_2\\ 0 & \gamma I-(M_1-M_2) & 0 & \cdots & 0\\ 0 & 0 & \ddots & \ddots & \vdots\\ \vdots & \vdots & \ddots & \ddots & 0\\ 0 & 0 & \cdots & 0 & \gamma I-(M_1-M_2)\end{bmatrix}\\
&=\det(\gamma I-(M_1+(n-1)M_2))(\det(\gamma I-(M_1-M_2)))^{n-1}
\end{aligned}$$

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$$
\begin{aligned}
&= \prod_{j=1}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \det\big[I_{\mathcal{N}} \otimes \big((\gamma - n_2\omega) I_{n_1} - \mathcal{L}_1\big) \\
&\quad -\omega^2\Big[\big(1_{\mathcal{N}} \otimes I_{n_1}\big)\big(1_{\mathcal{N}}^T \otimes I_{n_1}\big) \otimes \Big(1_{n_2}^T \big((\gamma - \mathcal{N}\omega) I_{n_2} - \mathcal{L}_2\big)^{-1} 1_{n_2}\Big)\Big] \\
&= \prod_{j=1}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \det\big[I_{\mathcal{N}} \otimes \big((\gamma - n_2\omega) I_{n_1} - \mathcal{L}_1\big) \\
&\quad - \omega^2 \Gamma_{\mathcal{L}_2}(\gamma - \mathcal{N}\omega)\big(1_{\mathcal{N}} 1_{\mathcal{N}}^T \otimes I_{n_1}\big)\big] \\
&= \prod_{j=1}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \det\big(\gamma I_{\mathcal{N}n_1} - M\big)
\end{aligned}
$$

where M is defined as Proposition 2 with $M_1 = (n_2\omega + \omega^2\Gamma_{\mathcal{L}_2}(\gamma - \mathcal{N}\omega))I_{n_1} + \mathcal{L}_1$ and $M_2 = \omega^2\Gamma_{\mathcal{L}_2}(\gamma - \mathcal{N}\omega)I_{n_1}$.

By computing, we can know that $M_1 + (\mathcal{N}-1)M_2 = (n_2\omega + \mathcal{N}\omega^2\Gamma_{\mathcal{L}_2}(\gamma - \mathcal{N}\omega))I_{n_1} + \mathcal{L}_1$ and $M_1 - M_2 = n_2\omega I_{n_1} + \mathcal{L}_1$.

From Propositions 2 and 3, we can have

$$
\begin{aligned}
&\Phi_{\mathcal{G}}(\mathcal{L} : \gamma) \\
&= \prod_{j=1}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \prod_{i=1}^{n_1}(\gamma - n_2\omega - \lambda_i)^{\mathcal{N}-1} \\
&\quad \prod_{i=1}^{n_1}\big(\gamma - n_2\omega - \mathcal{N}\omega^2\Gamma_{\mathcal{L}_2}(\gamma - \mathcal{N}\omega) - \lambda_i\big) \\
&= \prod_{j=1}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \prod_{i=1}^{n_1}(\gamma - n_2\omega - \lambda_i)^{\mathcal{N}-1} \\
&\quad \prod_{i=1}^{n_1}\left(\gamma - n_2\omega - \frac{\mathcal{N}\omega^2 n_2}{\gamma - \mathcal{N}\omega} - \lambda_i\right) \\
&= \prod_{j=2}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1}(\gamma - \mathcal{N}\omega - \mu_1)^{n_1} \\
&\quad \prod_{i=1}^{n_1}\frac{\gamma^2 - (\mathcal{N}\omega + n_2\omega + \lambda_i)\gamma + \mathcal{N}\omega\lambda_i}{\gamma - \mathcal{N}\omega} \prod_{i=1}^{n_1}(\gamma - n_2\omega - \lambda_i)^{\mathcal{N}-1} \\
&\overset{\mu_1=0}{=} \prod_{j=2}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1}(\gamma - \mathcal{N}\omega)^{n_1} \frac{1}{(\gamma - \mathcal{N}\omega)^{n_1}} \\
&\quad \prod_{i=1}^{n_1}\big(\gamma^2 - (\mathcal{N}\omega + n_2\omega + \lambda_i)\gamma + \mathcal{N}\omega\lambda_i\big) \prod_{i=1}^{n_1}(\gamma - n_2\omega - \lambda_i)^{\mathcal{N}-1} \\
&= \prod_{i=1}^{n_1}\big(\gamma^2 - (\mathcal{N}\omega + n_2\omega + \lambda_i)\gamma + \mathcal{N}\omega\lambda_i\big) \\
&\quad \prod_{j=2}^{n_2}(\gamma - \mathcal{N}\omega - \mu_j)^{n_1} \prod_{i=1}^{n_1}(\gamma - n_2\omega - \lambda_i)^{\mathcal{N}-1}
\end{aligned}
$$

as required. At the same time, we can immediately get $\sigma(\mathcal{G}) = \Omega_1 \cup \Omega_2 \cup \Omega_3$, where $\Omega_1 = \Omega_{11} \cup \Omega_{12}$ with $\Omega_{11} = \{\delta_{i1} | \delta_{i1} = ((\mathcal{N}\omega + n_2\omega + \lambda_i) + ((\mathcal{N}\omega + n_2\omega + \lambda_i)^2 - 4\mathcal{N}\omega\lambda_i)^{1/2})/2, i = 1, 2, \ldots, n_1\}$ and $\Omega_{12} = \{\delta_{i2} | \delta_{i2} = ((\mathcal{N}\omega + n_2\omega + \lambda_i) - ((\mathcal{N}\omega + n_2\omega + \lambda_i)^2 - 4\mathcal{N}\omega\lambda_i)^{1/2})/2, i = 1, 2, \ldots, n_1\}$; $\Omega_2 = \{\mu_j + \mathcal{N}\omega | j = 2, 3, \ldots, n_2\}$ and each eigenvalue in Ω_2 is n_1-repeated; and $\Omega_3 = \{\lambda_i + n_2\omega | i = 1, 2, \ldots, n_1\}$ and each eigenvalue in Ω_3 is $(\mathcal{N}-1)$-repeated.

From Lemma 1, some properties for $\mathcal{L}$-spectrum of $\mathcal{G}_1 \underline{\ast} \mathcal{G}_2$ can be obtained.

Proposition 4: The eigenvalues of $\mathcal{L}$ have the following properties.

1) $0 \in \Omega_1$.
2) $\mathcal{N}\omega \notin \Omega_1$.
3) $\Omega_{11} \cap \Omega_{12} = \emptyset$ and $\delta_{i1} > \mathcal{N}\omega$, $\delta_{i2} < \mathcal{N}\omega$.
4) $\Omega_2 \cap \Omega_{12} = \emptyset$.

Proof:

1) In Ω_1, we can have $\gamma^2 - (\mathcal{N}\omega + n_2\omega + \lambda_i)\gamma + \mathcal{N}\omega\lambda_i = \gamma^2 - (\mathcal{N}\omega + n_2\omega)\gamma = \gamma(\gamma - (\mathcal{N}\omega + n_2\omega))$ for $\lambda_1 = 0$, which yields $\gamma = 0 \in \Omega_1$, that is, 0 is the eigenvalue of $\mathcal{L}$, which is consistent with the fact that the Laplacian matrix has 0 eigenvalue.
2) If $\mathcal{N}\omega \in \Omega_1$, then $n_2 = 0$, which contradicts the fact $n_2 \neq 0$. Thus, $\mathcal{N}\omega \notin \Omega_1$.
3) For δ_{i1}, we have

$$
\begin{aligned}
\delta_{i1} &= \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) + \sqrt{(\mathcal{N}\omega + n_2\omega + \lambda_i)^2 - 4\mathcal{N}\omega\lambda_i}}{2} \\
&= \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) + \sqrt{(n_2\omega + \lambda_i - \mathcal{N}\omega)^2 + 4\mathcal{N}\omega^2 n_2}}{2} \\
&> \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) + \sqrt{(n_2\omega + \lambda_i - \mathcal{N}\omega)^2}}{2} \\
&= \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) + |n_2\omega + \lambda_i - \mathcal{N}\omega|}{2} \\
&= \begin{cases} n_2\omega + \lambda_i > \mathcal{N}\omega, & \text{if } \lambda_i > (\mathcal{N} - n_2)\omega \\ \mathcal{N}\omega, & \text{if } \lambda_i \leq (\mathcal{N} - n_2)\omega \end{cases}
\end{aligned}
$$

and thus, $\delta_{i1} > \mathcal{N}\omega$. Similarly,

$$
\begin{aligned}
\delta_{i2} &= \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) - \sqrt{(n_2\omega + \lambda_i - \mathcal{N}\omega)^2 + 4\mathcal{N}\omega^2 n_2}}{2} \\
&< \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) - \sqrt{(n_2\omega + \lambda_i - \mathcal{N}\omega)^2}}{2} \\
&= \frac{(\mathcal{N}\omega + n_2\omega + \lambda_i) - |n_2\omega + \lambda_i - \mathcal{N}\omega|}{2} \\
&= \begin{cases} \mathcal{N}\omega, & \text{if } \lambda_i \geq (\mathcal{N} - n_2)\omega \\ n_2\omega + \lambda_i < \mathcal{N}\omega, & \text{if } \lambda_i < (\mathcal{N} - n_2)\omega \end{cases}
\end{aligned}
$$

and thus, $\delta_{i2} < \mathcal{N}\omega$. Therefore, $\Omega_{11} \cap \Omega_{12} = \emptyset$.
4) Since $\delta_{i2} < \mathcal{N}\omega$ and $\mu_j + \mathcal{N}\omega > \mathcal{N}\omega$, obviously, $\Omega_2 \cap \Omega_{12} = \emptyset$.

□

C. Eigenvectors of the Laplacian Matrix $\mathcal{L}$ of $\mathcal{G}_1 \underline{\ast} \mathcal{G}_2$

In what follows, we will discuss the eigenvectors of the Laplacian matrix $\mathcal{L}$ of $\mathcal{G}_1 \underline{\ast} \mathcal{G}_2$ on account of Lemma 1.

Lemma 2: Let $\alpha_1, \alpha_2, \ldots, \alpha_{n_1}$ be the orthogonal eigenvectors of $\mathcal{L}_1$ corresponding to the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_{n_1}$ and $\beta_1, \beta_2, \ldots, \beta_{n_2}$ be the orthogonal eigenvectors of $\mathcal{L}_2$ corresponding to the eigenvalues $\mu_1, \mu_2, \ldots, \mu_{n_2}$. Then, the eigenvectors of the Laplacian matrix $\mathcal{L}$ of $\mathcal{G}_1 \underline{\ast} \mathcal{G}_2$ can be expressed as follows.

1)

$$
\xi_{i1} = \begin{bmatrix} 1_{\mathcal{N}} \otimes \alpha_i \\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2} \end{bmatrix}
$$

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and

$$\xi_{i2}=\begin{bmatrix}1_{\mathcal{N}}\otimes\alpha_i\\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega-\delta_{i2}}\otimes 1_{n_2}\end{bmatrix}$$

$(i=1,2,\ldots,n_1)$ corresponding to the eigenvalue δ_{i1} and δ_{i2} in Ω_1, respectively.

2)
$$\xi_{ij}=\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\beta_j\end{bmatrix}$$
$(i=1,2,\ldots,n_1$ and $j=2,\ldots,n_2)$ corresponding to the n_1 repeated eigenvalue $\mu_j+\mathcal{N}\omega$ in Ω_2, where $e_i\in\mathcal{R}^{n_1}$ indicates the unit vector.

3)
$$\xi_{li}=\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}$$
$(i=1,2,\ldots,n_1, l=1,2,\ldots,\mathcal{N}-1)$ corresponding to the $\mathcal{N}-1$ repeated eigenvalue $n_2\omega+\lambda_i$ in Ω_3, where $e_l\in\mathcal{R}^{\mathcal{N}-1}$ indicates the unit vector.

Proof:

1) Since $\mathcal{L}_1\alpha_i=\lambda_i\alpha_i$ and $\mathcal{L}_2 1_{n_2}=0$, we can have, as shown in the equation at the bottom of the next page. This implies that
$$\xi_{i1}=\begin{bmatrix}1_{\mathcal{N}}\otimes\alpha_i\\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega-\delta_{i1}}\otimes 1_{n_2}\end{bmatrix}$$
is the eigenvector corresponding to the eigenvalue δ_{i1} in Ω_{11} for $i=1,2,\ldots,n_1$. Similarly, we can prove that
$$\xi_{i2}=\begin{bmatrix}1_{\mathcal{N}}\otimes\alpha_i\\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega-\delta_{i2}}\otimes 1_{n_2}\end{bmatrix}$$
is the eigenvector corresponding to the eigenvalue δ_{i2} in Ω_{12} for $i=1,2,\ldots,n_1$. For simplicity, let $p_{i1}=\mathcal{N}\omega/(\mathcal{N}\omega-\delta_{i1})$ and $p_{i2}=\mathcal{N}\omega/(\mathcal{N}\omega-\delta_{i2})$, from Proposition 4 (3), it is easy to see $p_{i1}<0$, $p_{i2}>0$.

2) Since $\mathcal{L}_2\beta_j=\mu_j\beta_j$ and $1_{n_2}^T\beta_j=0$, we can have
$$\begin{aligned}&\mathcal{L}\xi_{ij}\\ &=\begin{bmatrix}I_{\mathcal{N}}\otimes\left(\mathcal{L}_1+n_2\omega I_{n_1}\right)-1_{\mathcal{N}}\otimes\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\\ -1_{\mathcal{N}}^T\otimes\left(\omega I_{n_1}\otimes 1_{n_2}\right)\ I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}\omega I_{n_2}\right)\end{bmatrix}\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\beta_j\end{bmatrix}\\ &=\begin{bmatrix}\mathcal{L}_1+n_2\omega I_{n_1} & 0 & 0 & -\omega I_{n_1}\otimes 1_{n_2}^T\\ 0 & \ddots & 0 & \vdots\\ 0 & 0 & \mathcal{L}_1+n_2\omega I_{n_1} & -\omega I_{n_1}\otimes 1_{n_2}^T\\ -\omega I_{n_1}\otimes 1_{n_2} & \cdots & -\omega I_{n_1}\otimes 1_{n_2} & I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}\omega I_{n_2}\right)\end{bmatrix}\\ &\quad\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\beta_j\end{bmatrix}\\ &=\begin{bmatrix}-\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\left(e_i\otimes\beta_j\right)\\ \vdots\\ -\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\left(e_i\otimes\beta_j\right)\\ \left(I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}I_{n_2}\right)\right)\left(e_i\otimes\beta_j\right)\end{bmatrix}\\ &=\begin{bmatrix}-1_{\mathcal{N}}\otimes\left(\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\left(e_i\otimes\beta_j\right)\right)\\ \left(I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}I_{n_2}\right)\right)\left(e_i\otimes\beta_j\right)\end{bmatrix}\\ &=\begin{bmatrix}-1_{\mathcal{N}}\otimes\left(\omega e_i\otimes\left(1_{n_2}^T\beta_j\right)\right)\\ e_i\otimes\left(\mathcal{L}_2\beta_j+\mathcal{N}\omega\beta_j\right)\end{bmatrix}\\ &=\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\left(\mu_j\beta_j+\mathcal{N}\omega\beta_j\right)\end{bmatrix}\\ &=\left(\mu_j+\mathcal{N}\omega\right)\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\beta_j\end{bmatrix}\\ &=\left(\mu_j+\mathcal{N}\omega\right)\xi_{ij}\end{aligned}$$
which implies that
$$\xi_{ij}=\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ e_i\otimes\beta_j\end{bmatrix}$$
is the eigenvector corresponding to the n_1 repeated eigenvalue $\mu_j+\mathcal{N}\omega$ in Ω_2 for $j=2,\ldots,n_2$.

3) Since $\mathcal{L}_1\alpha_i=\lambda_i\alpha_i$, we can have, as shown in the equation at the bottom of the next page, which implies that
$$\xi_{li}=\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}$$
is the eigenvector corresponding to the eigenvalue $\lambda_i+n_2\omega$ in Ω_3 for $i=1,2,\ldots,n_1$.

□

To sum up, we obtain $2n_1+n_1(\mathcal{N}-1)+n_1(n_2-1)=n_1(n_2+\mathcal{N})$ linear independent eigenvectors of $\mathcal{L}$ for $\mathcal{G}_1\underline{*}\mathcal{G}_2$.

D. Problem Statement

Here, the problem is briefly stated. The main goal is to find a necessary and sufficient algebra-based criterion for the controllability of the $\mathcal{N}$-duplication Corona product network (2) based on the Popov-Belevitch-Hautus (PBH) test. The Gram matrix criterion, the Kalman rank criterion, and the PBH test are three classical controllability criteria of mutual equivalence. The appearance of the Gram matrix criterion form often represents that related to the minimum control energy problem. The Kalman rank criterion is often used to simply judge and verify the controllability of a single and simple network. For instance, Example 1 will be exhibited by using the Kalman rank criterion in the example section later. Generally speaking, the PBH eigenvector criterion is mainly used in theoretical analysis, which further reveals the relationship between eigenvalues, eigenvectors, and controllability. In this article, the PBH test method is used to analyze the controllability of the entire composite MANs from the features of factor networks.

IV. Controllability Analysis

Next, we will discuss controllable conditions of the $\mathcal{N}$-duplication Corona product network (2) based on the eigenvectors of the $\mathcal{N}$-duplication Corona product network.

Proposition 5: (PBH Test) System $(\mathcal{G},\Sigma)$ is uncontrollable if and only if there is a left eigenvector ξ corresponding to eigenvalue λ of $\mathcal{L}$ such that $\xi^T\mathcal{B}=0$.

Theorem 1: As $\mathcal{N}>1$, the $\mathcal{N}$-duplication weighted Corona product network (2) is controllable if and only if the following conditions hold.

1) $(\mathcal{L}_1,\mathcal{B}_1)$ is controllable.
2) $(\mathcal{L}_2,\mathcal{B}_2)$ is controllable.
3) If there is a common eigenvalue $\eta=\Omega_{11}\cap\Omega_2\cap\Omega_3$, then $(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}^T+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}^T)\mathcal{B}_1\neq 0$ or

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$1_{\mathcal{N}-1}^T \otimes (\sum_{k=1}^{t} \rho_{i_k} \alpha_{i_k}^T \mathcal{B}_1) - \sum_{l'=1}^{\mathcal{N}-1} \sum_{k'=1}^{t'} \rho_{l'i'_{k'}} e_{l'}^T \otimes (\alpha_{i'_{k'}}^T \mathcal{B}_1) \neq 0$ or $\sum_{k=1}^{t} \rho_{i_k} p_{i_k 1} \alpha_{i_k}^T \otimes (1_{n_2}^T \mathcal{B}_2) + \sum_{h=1}^{s} \sum_{l=1}^{n_1} \rho_{l j_h} e_l^T \otimes (\beta_{j_h}^T \mathcal{B}_2) \neq 0$ holds, where $p_{i_k 1} = \mathcal{N}\omega/(\mathcal{N}\omega - \delta_{i_k 1})$ $(k = 1, 2, \ldots, t)$, $\eta = \delta_{i1} = \delta_{i_1 1} = \delta_{i_2 1} = \cdots = \delta_{i_t 1} = \mu_j + \mathcal{N}\omega = \mu_{j_1} + \mathcal{N}\omega = \mu_{j_2} + \mathcal{N}\omega = \cdots = \mu_{j_s} + \mathcal{N}\omega = \lambda_{i'} + n_2\omega = \lambda_{i'_1} + n_2\omega = \lambda_{i'_2} + n_2\omega = \cdots = \lambda_{i'_{t'}} + n_2\omega$ (δ_{i1} with multiplicity-t, $\mu_j + \mathcal{N}\omega$ with multiplicity-$n_1 s$, and $\lambda_{i'} + n_2\omega$ with multiplicity-$(\mathcal{N} - 1)t'$), and arbitrary constants $\rho_{i_1}, \ldots, \rho_{i_t}, \rho_{1j_1}, \ldots, \rho_{1j_s}, \ldots, \rho_{n_1 j_1}, \ldots, \rho_{n_1 j_s}, \rho_{1i'_1}, \ldots, \rho_{1i'_{t'}}, \ldots, \rho_{(\mathcal{N}-1)i'_1}, \ldots, \rho_{(\mathcal{N}-1)i'_{t'}}$ are not all zero.

Proof: Lemma 1 gives three kinds of eigenvalues for $\mathcal{N}$-duplication weighted Corona product network (2)—single eigenvalues in Ω_1, n_1-repeated eigenvalues in Ω_2, and $(\mathcal{N} - 1)$-repeated eigenvalues in Ω_3. Next, for the convenience of discussion, we will discuss eigenvalues in Ω_2, Ω_3, and Ω_1 based on Propositions 4 and 5 and Lemmas 1 and 2. □

Sufficiency (Case 1): Consider the eigenvalues in Ω_2. First, if the eigenvalue μ_j of $\mathcal{L}_2$ is single

$$\xi_{ij} = \begin{bmatrix} 0_{\mathcal{N}n_1 \times 1} \\ e_i \otimes \beta_j \end{bmatrix}$$

is the eigenvector corresponding to the eigenvalue $\mu_j + \mathcal{N}\omega$ with multiplicity-n_1 in Ω_2, and the linear combination

$$\xi_j = \begin{bmatrix} 0_{\mathcal{N}n_1 \times 1} \\ \sum_{i=1}^{n_1} \rho_i e_i \otimes \beta_j \end{bmatrix}$$

is still the eigenvector corresponding to the eigenvalue $\mu_j + \mathcal{N}\omega$ for $i = 1, 2, \ldots, n_1$ and $j = 2, \ldots, n_2$, where arbitrary constants $\rho_1, \rho_2, \ldots, \rho_{n_1}$ are not all zero. Thus,

$$\begin{aligned} &\xi_j^T \mathcal{B} \\ &= \begin{bmatrix} 0_{\mathcal{N}n_1 \times 1} \\ \sum_{i=1}^{n_1} \rho_i e_i \otimes \beta_j \end{bmatrix}^T \mathcal{B} \\ &= \begin{bmatrix} 0_{\mathcal{N}n_1 \times 1} \\ \sum_{i=1}^{n_1} \rho_i e_i \otimes \beta_j \end{bmatrix}^T \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
&\mathcal{L}\xi_{i1} \\
&= \begin{bmatrix} I_{\mathcal{N}} \otimes (\mathcal{L}_1 + n_2\omega I_{n_1}) & -1_{\mathcal{N}} \otimes (\omega I_{n_1} \otimes 1_{n_2}^T) \\ -1_{\mathcal{N}}^T \otimes (\omega I_{n_1} \otimes 1_{n_2}) & I_{n_1} \otimes (\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}) \end{bmatrix} \begin{bmatrix} 1_{\mathcal{N}} \otimes \alpha_i \\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2} \end{bmatrix} \\
&= \begin{bmatrix} \mathcal{L}_1 + n_2\omega I_{n_1} & 0 & 0 & -\omega I_{n_1} \otimes 1_{n_2}^T \\ 0 & \ddots & 0 & \vdots \\ 0 & 0 & \mathcal{L}_1 + n_2\omega I_{n_1} & -\omega I_{n_1} \otimes 1_{n_2}^T \\ -\omega I_{n_1} \otimes 1_{n_2} & \cdots & -\omega I_{n_1} \otimes 1_{n_2} & I_{n_1} \otimes (\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}) \end{bmatrix} \\
&\quad \begin{bmatrix} \alpha_i \\ \vdots \\ \alpha_i \\ \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2} \end{bmatrix} \\
&= \begin{bmatrix} \left((\mathcal{L}_1 + n_2\omega I_{n_1})\alpha_i - (\omega I_{n_1} \otimes 1_{n_2}^T)\left(\frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}\right)\right) \\ \vdots \\ \left((\mathcal{L}_1 + n_2\omega I_{n_1})\alpha_i - (\omega I_{n_1} \otimes 1_{n_2}^T)\left(\frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}\right)\right) \\ -1_{\mathcal{N}}^T 1_{\mathcal{N}} \otimes (\omega I_{n_1} \otimes 1_{n_2})\alpha_i + (I_{n_1} \otimes (\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}))\left(\frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}\right) \end{bmatrix} \\
&= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left((\mathcal{L}_1 + n_2\omega I_{n_1})\alpha_i - (\omega I_{n_1} \otimes 1_{n_2}^T)\left(\frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}\right)\right) \\ -1_{\mathcal{N}}^T 1_{\mathcal{N}} \otimes (\omega I_{n_1} \otimes 1_{n_2})\alpha_i + (I_{n_1} \otimes (\mathcal{L}_2 + \mathcal{N}\omega I_{n_2}))\left(\frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}\right) \end{bmatrix} \\
&= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\lambda_i\alpha_i + n_2\omega\alpha_i - \frac{\mathcal{N}\omega^2\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes 1_{n_2}^T 1_{n_2}\right) \\ -\mathcal{N}\omega(I_{n_1} \otimes 1_{n_2})(\alpha_i \otimes 1) + \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes ((\mathcal{L}_2 + \mathcal{N}\omega I_{n_2})1_{n_2}) \end{bmatrix} \\
&= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\lambda_i\alpha_i + n_2\omega\alpha_i - n_2\frac{\mathcal{N}\omega^2\alpha_i}{\mathcal{N}\omega - \delta_{i1}}\right) \\ -\mathcal{N}\omega(\alpha_i \otimes 1_{n_2}) + \frac{\mathcal{N}\omega\alpha_i}{\mathcal{N}\omega - \delta_{i1}} \otimes (\mathcal{L}_2 1_{n_2} + \mathcal{N}\omega 1_{n_2}) \end{bmatrix} \\
&= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\left(\lambda_i + n_2\left(\omega - \frac{\mathcal{N}\omega^2}{\mathcal{N}\omega - \delta_{i1}}\right)\right)\alpha_i\right) \\ -\mathcal{N}\omega(\alpha_i \otimes 1_{n_2}) + \frac{\mathcal{N}^2\omega^2}{\mathcal{N}\omega - \delta_{i1}}(\alpha_i \otimes 1_{n_2}) \end{bmatrix} \\
&= \begin{bmatrix} \delta_{i1} 1_{\mathcal{N}} \otimes \alpha_i \\ \delta_{i1} \frac{\mathcal{N}\omega}{\mathcal{N}\omega - \delta_{i1}} \alpha_i \otimes 1_{n_2} \end{bmatrix} \\
&= \delta_{i1}\xi_{i1}
\end{aligned}$$

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$$
\begin{aligned}
&= \left[0_{1\times\mathcal{N}n_1}\ \left(\sum_{i=1}^{n_1}\rho_i e_i\otimes\beta_j\right)^T\left(I_{n_1}\otimes\mathcal{B}_2\right)\right]\\
&= \left[0_{1\times\mathcal{N}n_1}\ \sum_{i=1}^{n_1}\rho_i e_i^T\otimes\left(\beta_j^T\mathcal{B}_2\right)\right]\\
&= \left[0_{1\times\mathcal{N}n_1}\ \rho_1\beta_j^T\mathcal{B}_2\ \rho_2\beta_j^T\mathcal{B}_2\ \cdots\ \rho_{n_1}\beta_j^T\mathcal{B}_2\right]. \qquad (3)
\end{aligned}
$$

Because $(\mathcal{L}_2,\mathcal{B}_2)$ is controllable, from Proposition 5, we can have $\mathcal{B}_2\neq 0$ and $\beta_j^T\mathcal{B}_2\neq 0$, and thus, $\rho_i\beta_j^T\mathcal{B}_2\neq 0$, which implies that we cannot find constants $\rho_1,\ldots,\rho_{n_1}$ (not all zero) such that $\xi_j^T\mathcal{B}=0$.

Second, if $\mu_j=\mu_{j_1}=\mu_{j_2}=\cdots=\mu_{j_s}$, $\xi_s=\sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}\xi_{j_k}$ is the eigenvector corresponding to the eigenvalue $\mu_j+\mathcal{N}\omega$ with multiplicity-$n_1 s$ in Ω_2, where arbitrary constants $\rho_{1j_1},\ldots,\rho_{1j_s},\ldots,\rho_{n_1j_1},\ldots,\rho_{n_1j_s}$ are not all zero and $\xi_{j_1},\xi_{j_2},\ldots,\xi_{j_s}$ are the linearly independent eigenvectors corresponding to $\mu_{j_1}+\mathcal{N}\omega,\mu_{j_2}+\mathcal{N}\omega,\ldots,\mu_{j_s}+\mathcal{N}\omega$. Thus,

$$
\begin{aligned}
&\xi_s^T\mathcal{B}\\
&= \begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ \sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}e_i\otimes\beta_{j_k}\end{bmatrix}^T\mathcal{B}\\
&= \begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ \sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}e_i\otimes\beta_{j_k}\end{bmatrix}^T\begin{bmatrix}I_{\mathcal{N}}\otimes\mathcal{B}_1 & 0\\ 0 & I_{n_1}\otimes\mathcal{B}_2\end{bmatrix}\\
&= \left[0_{1\times\mathcal{N}n_1}\ \left(\sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}e_i\otimes\beta_{j_k}\right)^T\left(I_{n_1}\otimes\mathcal{B}_2\right)\right]\\
&= \left[0_{1\times\mathcal{N}n_1}\ \sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}e_i^T\otimes\left(\beta_{j_k}^T\mathcal{B}_2\right)\right]\\
&= \left[0_{1\times\mathcal{N}n_1}\ \sum_{k=1}^{s}\rho_{1j_k}\beta_{j_k}^T\mathcal{B}_2\ \cdots\ \sum_{k=1}^{s}\rho_{n_1j_k}\beta_{j_k}^T\mathcal{B}_2\right]. \qquad (4)
\end{aligned}
$$

Because $(\mathcal{L}_2,\mathcal{B}_2)$ is controllable, from Proposition 5, we can have $\mathcal{B}_2\neq 0$ and $\beta_{j_k}^T\mathcal{B}_2\neq 0$, and thus, $\sum_{k=1}^{s}\rho_{ij_k}\beta_{j_k}^T\mathcal{B}_2\neq 0$, which implies that we cannot find constants $\rho_{1j_1},\ldots,\rho_{1j_s},\ldots,\rho_{n_1j_1},\ldots,\rho_{n_1j_s}$ (not all zero) such that $\xi_s^T\mathcal{B}=0$.

Case 2: Consider the eigenvalues in Ω_3. First, if the eigenvalue λ_i of $\mathcal{L}_1$ is single

$$
\xi_{li}=\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}
$$

is the eigenvector corresponding to the eigenvalue $\lambda_i+n_2\omega$ with multiplicity-$(\mathcal{N}-1)$ in Ω_3, and the linear combination

$$
\xi_i=\begin{bmatrix}\sum_{l=1}^{\mathcal{N}-1}\rho_l\alpha_i\\ -\sum_{l=1}^{\mathcal{N}-1}\rho_l e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}
$$

is still the eigenvector corresponding to the eigenvalue $\lambda_i+n_2\omega$ for $l=1,2,\ldots,\mathcal{N}-1$ and $i=1,2,\ldots,n_1$, where arbitrary constants $\rho_1,\rho_2,\ldots,\rho_{(\mathcal{N}-1)}$ are not all zero. Thus,

$$
\begin{aligned}
&\xi_i^T\mathcal{B}\\
&= \begin{bmatrix}\sum_{l=1}^{\mathcal{N}-1}\rho_l\alpha_i\\ -\sum_{l=1}^{\mathcal{N}-1}\rho_l e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}^T\mathcal{B}\\
&= \begin{bmatrix}\sum_{l=1}^{\mathcal{N}-1}\rho_l\alpha_i\\ -\sum_{l=1}^{\mathcal{N}-1}\rho_l e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}^T\begin{bmatrix}I_{\mathcal{N}}\otimes\mathcal{B}_1 & 0\\ 0 & I_{n_1}\otimes\mathcal{B}_2\end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&\mathcal{L}\xi_{li}\\
&= \begin{bmatrix}I_{\mathcal{N}}\otimes\left(\mathcal{L}_1+n_2\omega I_{n_1}\right) & -1_{\mathcal{N}}\otimes\left(\omega I_{n_1}\otimes 1_{n_2}^T\right)\\ -1_{\mathcal{N}}^T\otimes\left(\omega I_{n_1}\otimes 1_{n_2}\right) & I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}\omega I_{n_2}\right)\end{bmatrix}\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= \begin{bmatrix}\mathcal{L}_1+n_2\omega I_{n_1} & 0 & -\omega I_{n_1}\otimes 1_{n_2}^T\\ 0 & I_{\mathcal{N}-1}\otimes\left(\mathcal{L}_1+n_2\omega I_{n_1}\right) & -1_{\mathcal{N}-1}\otimes\left(\omega I_{n_2}\otimes 1_{n_2}^T\right)\\ -\omega I_{n_1}\otimes 1_{n_2} & -1_{\mathcal{N}-1}^T\otimes\left(\omega I_{n_1}\otimes 1_{n_2}\right) & I_{n_1}\otimes\left(\mathcal{L}_2+\mathcal{N}\omega I_{n_2}\right)\end{bmatrix}\\
&\quad\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= \begin{bmatrix}\left(\mathcal{L}_1+n_2\omega I_{n_1}\right)\alpha_i\\ \left(I_{\mathcal{N}-1}\otimes\left(\mathcal{L}_1+n_2\omega I_{n_1}\right)\right)(-e_l\otimes\alpha_i)\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= \begin{bmatrix}\left(\mathcal{L}_1+n_2\omega I_{n_1}\right)\alpha_i\\ -e_l\otimes\left(\left(\mathcal{L}_1+n_2\omega I_{n_1}\right)\alpha_i\right)\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= \begin{bmatrix}\lambda_i\alpha_i+n_2\omega\alpha_i\\ -e_l\otimes\left(\lambda_i\alpha_i+n_2\omega\alpha_i\right)\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= (\lambda_i+n_2\omega)\begin{bmatrix}\alpha_i\\ -e_l\otimes\alpha_i\\ 0_{n_1n_2\times 1}\end{bmatrix}\\
&= (\lambda_i+n_2\omega)\xi_{li}
\end{aligned}
$$

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$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \rho_l \alpha_i \\ -\sum_{l=1}^{\mathcal{N}-1} \rho_l e_l \otimes \alpha_i \\ 0_{n_1 n_2 \times 1} \end{bmatrix}^T \begin{bmatrix} \mathcal{B}_1 & 0 & 0 \\ 0 & I_{\mathcal{N}-1} \otimes \mathcal{B}_1 & 0 \\ 0 & 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix}$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \rho_l \alpha_i^T \mathcal{B}_1 \ -\sum_{l=1}^{\mathcal{N}-1} \rho_l e_l^T \otimes \left(\alpha_i^T \mathcal{B}_1\right) \ 0_{1 \times n_1 n_2} \right]$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \rho_l \alpha_i^T \mathcal{B}_1 \ -\rho_1 \alpha_i^T \mathcal{B}_1 \ \cdots \ -\rho_{(\mathcal{N}-1)} \alpha_i^T \mathcal{B}_1 \ 0_{1 \times n_1 n_2} \right]. \quad (5)$$

Because $(\mathcal{L}_1, \mathcal{B}_1)$ is controllable, from Proposition 5, we can have $\alpha_i^T \mathcal{B}_1 \neq 0$, and thus, $\rho_l \alpha_i^T \mathcal{B}_1 \neq 0$, which implies that we cannot find constants $\rho_1, \rho_2, \ldots, \rho_{(\mathcal{N}-1)}$ (not all zero) such that $\xi_i^T \mathcal{B} = 0$.

Second, if $\lambda_i = \lambda_{i_1} = \lambda_{i_2} = \cdots = \lambda_{i_t}$

$$\xi_t = \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ 0_{n_1 n_2 \times 1} \end{bmatrix}$$

is the eigenvector corresponding to the eigenvalue $\lambda_i + n_2\omega$ with multiplicity-$(\mathcal{N}-1)t$ in Ω_3 for $i = 1, 2, \ldots, n_1$, where arbitrary constants $\rho_{1 i_1}, \ldots, \rho_{1 i_t}, \ldots, \rho_{(\mathcal{N}-1) i_1}, \ldots, \rho_{(\mathcal{N}-1) i_t}$ are not all zero. Thus,

$$\xi_t^T \mathcal{B}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ 0_{n_1 n_2 \times 1} \end{bmatrix}^T \mathcal{B}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ 0_{n_1 n_2 \times 1} \end{bmatrix}^T \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ 0_{n_1 n_2 \times 1} \end{bmatrix}^T \begin{bmatrix} \mathcal{B}_1 & 0 & 0 \\ 0 & I_{\mathcal{N}-1} \otimes \mathcal{B}_1 & 0 \\ 0 & 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix}$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 \quad \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l^T \otimes \left(\alpha_{i_k}^T \mathcal{B}_1\right) \right]$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 - \sum_{k=1}^{t} \rho_{1 i_k} \alpha_{i_k}^T \mathcal{B}_1 \cdots \right.$$
$$\left. - \sum_{k=1}^{t} \rho_{(\mathcal{N}-1) i_k} \alpha_{i_k}^T \mathcal{B}_1 \ 0_{1 \times n_1 n_2} \right]. \quad (6)$$

Because $(\mathcal{L}_1, \mathcal{B}_1)$ is controllable, from Proposition 5, $\sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 \neq 0$, which implies that we cannot find constants $\rho_{1 i_1}, \ldots, \rho_{1 i_t}, \ldots, \rho_{(\mathcal{N}-1) i_1}, \ldots, \rho_{(\mathcal{N}-1) i_t}$ (not all zero) such that $\xi_t^T \mathcal{B} = 0$.

Third, if $\Omega_2 \cap \Omega_3 \neq \emptyset$, let $\theta \in \Omega_2 \cap \Omega_3$, then

$$\xi_{ts} = \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ \sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i \otimes \beta_{j_k} \end{bmatrix}$$

is the eigenvector corresponding to the eigenvalue $\theta = \mu_j + \mathcal{N}\omega = \mu_{j_1} + \mathcal{N}\omega = \cdots = \mu_{j_s} + \mathcal{N}\omega = \lambda_{i_1} + n_2\omega = \lambda_{i_t} + n_2\omega = \lambda_i + n_2\omega$ ($\mu_j + \mathcal{N}\omega$ with multiplicity-$n_1 s$ and $\lambda_i + n_2\omega$ with multiplicity-$(\mathcal{N}-1)t$) for $i = 1, 2, \ldots, n_1$ and $j = 2, \ldots, n_2$, where arbitrary constants $\rho_{1 i_1}, \ldots, \rho_{1 i_s}, \ldots, \rho_{(\mathcal{N}-1) i_1}, \ldots, \rho_{(\mathcal{N}-1) i_s}, \rho_{1 j_1}, \ldots, \rho_{n_1 j_s}$ are not all zero. Thus,

$$\xi_{ts}^T \mathcal{B}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ \sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i \otimes \beta_{j_k} \end{bmatrix}^T \mathcal{B}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ \sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i \otimes \beta_{j_k} \end{bmatrix}^T \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix}$$
$$= \begin{bmatrix} \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k} \\ -\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \\ \sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i \otimes \beta_{j_k} \end{bmatrix}^T$$
$$\begin{bmatrix} \mathcal{B}_1 & 0 & 0 \\ 0 & I_{\mathcal{N}-1} \otimes \mathcal{B}_1 & 0 \\ 0 & 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix}$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 \right.$$
$$\left(-\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l \otimes \alpha_{i_k} \right)^T (I_{\mathcal{N}-1} \otimes \mathcal{B}_1)$$
$$\left. \left(\sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i \otimes \beta_{j_k} \right)^T \left(I_{n_1} \otimes \mathcal{B}_2\right) \right]$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 \quad - \sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} e_l^T \otimes \left(\alpha_{i_k}^T \mathcal{B}_1\right) \right.$$
$$\left. \sum_{k=1}^{s} \sum_{i=1}^{n_1} \rho_{i j_k} e_i^T \otimes \left(\beta_{j_k}^T \mathcal{B}_2\right) \right]$$
$$= \left[\sum_{l=1}^{\mathcal{N}-1} \sum_{k=1}^{t} \rho_{l i_k} \alpha_{i_k}^T \mathcal{B}_1 \right.$$
$$- \sum_{k=1}^{t} \rho_{1 i_k} \alpha_{i_k}^T \mathcal{B}_1 \ \cdots \ - \sum_{k=1}^{t} \rho_{(\mathcal{N}-1) i_k} \alpha_{i_k}^T \mathcal{B}_1$$
$$\left. \sum_{k=1}^{s} \rho_{1 j_k} \beta_{j_k}^T \mathcal{B}_2 \ \cdots \ \sum_{k=1}^{s} \rho_{n_1 j_k} \beta_{j_k}^T \mathcal{B}_2 \right]. \quad (7)$$

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Because $(\mathcal{L}_1, \mathcal{B}_1)$ and $(\mathcal{L}_2, \mathcal{B}_2)$ are both controllable, from Proposition 5, we can have $\alpha_{i_k}^T \mathcal{B}_1 \neq 0$ and $\beta_{j_k}^T \mathcal{B}_2 \neq 0$, and thus, $\xi_{ts}^T \mathcal{B} \neq 0$, which implies that we cannot find constants $\rho_{1i_1}, \ldots, \rho_{(\mathcal{N}-1)i_t}, \rho_{1j_1}, \ldots, \rho_{n_1 j_s}$ (not all zero) such that $\xi_{ts}^T \mathcal{B} \neq 0$.

Case 3: Consider the eigenvalues in Ω_1. First, if the eigenvalue δ_{iq} in Ω_{1q} is single

$$\xi_{iq} = \begin{bmatrix} 1_{\mathcal{N}} \otimes \alpha_i \\ p_{iq}\alpha_i \otimes 1_{n_2} \end{bmatrix}$$

is the eigenvector corresponding to the eigenvalue δ_{iq} for $i = 1, 2, \ldots, n_1$ and $q = 1, 2$. Thus,

$$\begin{aligned} \xi_{iq}^T \mathcal{B} &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \alpha_i \\ p_{iq}\alpha_i \otimes 1_{n_2} \end{bmatrix}^T \mathcal{B} \\ &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \alpha_i \\ p_{iq}\alpha_i \otimes 1_{n_2} \end{bmatrix}^T \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix} \\ &= \left[1_{\mathcal{N}}^T \otimes \left(\alpha_i^T \mathcal{B}_1\right) \; p_{iq}\alpha_i^T \otimes \left(1_{n_2}^T \mathcal{B}_2\right)\right]. \end{aligned} \tag{8}$$

Because $(\mathcal{L}_2, \mathcal{B}_2)$ is controllable, $\mathcal{B}_2 \neq 0$, and thus, $1_{n_2}^T \mathcal{B}_2 \neq 0$. From Proposition 4, we can know $p_{iq} \neq 0$ [see (3)]. Therefore, $p_{iq}\alpha_i^T \otimes (1_{n_2}^T \mathcal{B}_2) \neq 0$ ($i = 1, 2, \ldots, n_1$ and $q = 1, 2$) always holds. Thus, $\xi_{iq}^T \mathcal{B} \neq 0$.

Second, if $\lambda_i = \lambda_{i_1} = \lambda_{i_2} = \cdots = \lambda_{i_t}$, then $\delta_{iq} = \delta_{i_1 q} = \delta_{i_2 q} = \cdots = \delta_{i_t q}$ ($q = 1, 2$), and thus, $\xi_{tq} = \begin{bmatrix} 1_{\mathcal{N}} \otimes (\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k} \otimes 1_{n_2} \end{bmatrix}$ is the eigenvector corresponding to the eigenvalue δ_{i1} (or δ_{i2}) with multiplicity-t in Ω_{11} (or Ω_{12}) for $i = 1, 2, \ldots, n_1$, where arbitrary constants $\rho_{i_1}, \ldots, \rho_{i_t}$ are not all zero. Thus,

$$\begin{aligned} \xi_{tq}^T \mathcal{B} &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k} \otimes 1_{n_2} \end{bmatrix}^T \mathcal{B} \\ &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k} \otimes 1_{n_2} \end{bmatrix}^T \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix} \\ &= \left[1_{\mathcal{N}}^T \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}^T \mathcal{B}_1\right) \; \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k}^T \otimes \left(1_{n_2}^T \mathcal{B}_2\right)\right]. \end{aligned} \tag{9}$$

Because $(\mathcal{L}_2, \mathcal{B}_2)$ is controllable, $\mathcal{B}_2 \neq 0$ and $1_{n_2}^T \mathcal{B}_2 \neq 0$. From Proposition 4, we can know $p_{i_k q} \neq 0$ [see (3)], and then, $\sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k}^T \neq 0$; thus, $\sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k}^T \otimes (1_{n_2}^T \mathcal{B}_2) \neq 0$, which implies that we cannot find constants $\rho_{i_1}, \ldots, \rho_{i_t}$ (not all zero) such that $\xi_{tq}^T \mathcal{B} = 0$.

Third, if $\tau = \Omega_{11} \cap \Omega_2$, then

$$\xi_{\tau t} = \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k} \otimes 1_{n_2} + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i \otimes \beta_{j_k} \end{bmatrix}$$

is the eigenvector corresponding to the common eigenvalue $\tau = \delta_{i1} = \delta_{i_1 1} = \delta_{i_2 1} = \cdots = \delta_{i_t 1} = \mu_{j_1} + \mathcal{N}\omega = \mu_{j_2} + \mathcal{N}\omega = \cdots = \mu_{j_s} + \mathcal{N}\omega = \mu_j + \mathcal{N}\omega$ (δ_{i1} with multiplicity-t and $\mu_j + \mathcal{N}\omega$ with multiplicity-$n_1 s$) for $i = 1, 2, \ldots, n_1$ and $j = 2, \ldots, n_2$, where arbitrary constants $\rho_{i_1}, \ldots, \rho_{i_t}, \rho_{1j_1}, \ldots, \rho_{1j_s}, \ldots, \rho_{n_1 j_1}, \ldots, \rho_{n_1 j_s}$ are not all zero. Thus,

$$\begin{aligned} &\xi_{\tau t}^T \mathcal{B} \\ &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k} \otimes 1_{n_2} + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i \otimes \beta_{j_k} \end{bmatrix}^T \mathcal{B} \\ &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k} \otimes 1_{n_2} + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i \otimes \beta_{j_k} \end{bmatrix}^T \\ &\quad \begin{bmatrix} I_{\mathcal{N}} \otimes \mathcal{B}_1 & 0 \\ 0 & I_{n_1} \otimes \mathcal{B}_2 \end{bmatrix} \\ &= \left[\left(1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right)\right)^T (I_{\mathcal{N}} \otimes \mathcal{B}_1) \quad \left(\sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k} \otimes 1_{n_2} \right.\right. \\ &\qquad \left.\left. + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i \otimes \beta_{j_k}\right)^T \left(I_{n_1} \otimes \mathcal{B}_2\right)\right] \\ &= \left[1_{\mathcal{N}}^T \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}^T \mathcal{B}_1\right) \quad \left(\sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k} \otimes 1_{n_2} \right.\right. \\ &\qquad \left.\left. + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i \otimes \beta_{j_k}\right)^T \left(I_{n_1} \otimes \mathcal{B}_2\right)\right] \\ &= \left[1_{\mathcal{N}}^T \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}^T \mathcal{B}_1\right) \quad \left(\sum_{k=1}^t \rho_{i_k} p_{i_k 1}\alpha_{i_k}^T \otimes \left(1_{n_2}^T \mathcal{B}_2\right) \right.\right. \\ &\qquad \left.\left. + \sum_{k=1}^s \sum_{i=1}^{n_1} \rho_{ij_k} e_i^T \otimes \left(\beta_{j_k}^T \mathcal{B}_2\right)\right)\right]. \end{aligned} \tag{10}$$

Because $(\mathcal{L}_1, \mathcal{B}_1)$ and $(\mathcal{L}_2, \mathcal{B}_2)$ are controllable, $\alpha_{i_k}^T \mathcal{B}_1 \neq 0$ and $\beta_{j_k}^T \mathcal{B}_2 \neq 0$, and thus, $\xi_{\tau t}^T \mathcal{B} \neq 0$, which implies that we cannot find constants $\rho_{i_1}, \ldots, \rho_{i_t}, \rho_{1j_1}, \ldots, \rho_{1j_s}, \rho_{n_1 j_1}, \ldots, \rho_{n_1 j_s}$ (not all zero) such that $\xi_{\tau t}^T \mathcal{B} = 0$.

Fourth, if $\zeta_q = \Omega_{1q} \cap \Omega_3$ ($q = 1, 2$), then

$$\begin{aligned} &\xi_{\zeta_q t t'} \\ &= \begin{bmatrix} 1_{\mathcal{N}} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) \\ \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k} \otimes 1_{n_2} \end{bmatrix}^T \\ &\quad + \begin{bmatrix} \sum_{l'=1}^{\mathcal{N}-1} \sum_{k'=1}^{t'} \rho_{l' i'_{k'}}\alpha_{i'_{k'}} \\ -\sum_{l'=1}^{\mathcal{N}-1} \sum_{k'=1}^{t'} \rho_{l' i'_{k'}} e_{l'} \otimes \alpha_{i'_{k'}} \\ 0_{n_1 n_2 \times 1} \end{bmatrix}^T \\ &= \begin{bmatrix} \sum_{k=1}^t \rho_{i_k}\alpha_{i_k} + \sum_{l'=1}^{\mathcal{N}-1} \sum_{k'=1}^{t'} \rho_{l' i'_{k'}}\alpha_{i'_{k'}} \\ 1_{\mathcal{N}-1} \otimes \left(\sum_{k=1}^t \rho_{i_k}\alpha_{i_k}\right) - \sum_{l'=1}^{\mathcal{N}-1} \sum_{k'=1}^{t'} \rho_{l' i'_{k'}} e_{l'} \otimes \alpha_{i'_{k'}} \\ \sum_{k=1}^t \rho_{i_k} p_{i_k q}\alpha_{i_k} \otimes 1_{n_2} \end{bmatrix}^T \end{aligned}$$

is the eigenvector corresponding to the common eigenvalue $\zeta_q = \delta_{iq} = \delta_{i_1 q} = \delta_{i_2 q} = \cdots = \delta_{i_t q} = \lambda_{i'} + n_2\omega = \lambda_{i'_1} + n_2\omega =$

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$\lambda_{i'_2}+n_2\omega=\cdots=\lambda_{i'_{t'}}+n_2\omega$ (δ_{iq} with multiplicity-t and $\lambda_{i'}+n_2\omega$ with multiplicity-$(\mathcal{N}-1)t'$) for $i=1,2,\ldots,n_1$ and $i'=1,2,\ldots,n_1$, where $p_{i_kq}\neq 0$ and the arbitrary constants $\rho_{i_1},\ldots,\rho_{i_t},\rho_{1i'_1},\ldots,\rho_{1i'_{t'}},\ldots,\rho_{(\mathcal{N}-1)i'_1},\ldots,\rho_{(\mathcal{N}-1)i'_{t'}}$ are not all zero. Thus,

$$
\begin{aligned}
&\xi^T_{\zeta_q tt'}\mathcal{B}\\
&=\begin{bmatrix}\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\\ 1_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\\ \sum_{k=1}^{t}\rho_{i_k}p_{i_kq}\alpha_{i_k}\otimes 1_{n_2}\end{bmatrix}^T\\
&\quad\begin{bmatrix}\mathcal{B}_1 & 0 & 0\\ 0 & I_{\mathcal{N}-1}\otimes\mathcal{B}_1 & 0\\ 0 & 0 & I_{n_1}\otimes\mathcal{B}_2\end{bmatrix}\\
&=\left[\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\right)^T\mathcal{B}_1\right.\\
&\quad\left(1_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\right)^T\\
&\quad\left.(I_{\mathcal{N}-1}\otimes\mathcal{B}_1)\left(\sum_{k=1}^{t}\rho_{i_k}p_{i_kq}\alpha_{i_k}\otimes 1_{n_2}\right)^T(I_{n_1}\otimes\mathcal{B}_2)\right]\\
&=\left[\left(\sum_{k=1}^{t}\rho_{i_k}\alpha^T_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha^T_{i'_{k'}}\right)\mathcal{B}_1\right.\\
&\quad 1^T_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha^T_{i_k}\mathcal{B}_1\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e^T_{l'}\otimes\\
&\quad\left.\left(\alpha^T_{i'_{k'}}\mathcal{B}_1\right)\sum_{k=1}^{t}\rho_{i_k}p_{i_kq}\alpha^T_{i_k}\otimes\left(1^T_{n_2}\mathcal{B}_2\right)\right].
\end{aligned}
\tag{11}
$$

Because $(\mathcal{L}_1,\mathcal{B}_1)$ and $(\mathcal{L}_2,\mathcal{B}_2)$ are controllable, we can have $\xi^T_{\zeta_q tt'}\mathcal{B}\neq 0$, which implies that we cannot find constants $\rho_{i_1},\ldots,\rho_{i_t},\rho_{1i'_1},\ldots,\rho_{1i'_{t'}},\ldots,\rho_{(\mathcal{N}-1)i'_1},\ldots,\rho_{(\mathcal{N}-1)i'_{t'}}$(not all zero) such that $\xi^T_{\zeta_q tt'}\mathcal{B}=0$.

Fifth, if $\eta=\Omega_{11}\cap\Omega_2\cap\Omega_3$, then

$$
\begin{aligned}
&\xi_{\eta tst'}\\
&=\begin{bmatrix}1_{\mathcal{N}}\otimes\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}\\ \sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}\otimes 1_{n_2}\end{bmatrix}^T+\begin{bmatrix}0_{\mathcal{N}n_1\times 1}\\ \sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e_l\otimes\beta_{j_h}\end{bmatrix}^T\\
&\quad+\begin{bmatrix}\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\\ -\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\\ 0_{n_1n_2\times 1}\end{bmatrix}^T\\
&=\begin{bmatrix}\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\\ 1_{\mathcal{N}-1}\otimes\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\\ \sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}\otimes 1_{n_2}+\sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e_l\otimes\beta_{j_h}\end{bmatrix}^T
\end{aligned}
$$

is the eigenvector corresponding to the common eigenvalue $\eta=\delta_{i1}=\delta_{i_11}=\delta_{i_21}=\cdots=\delta_{i_t1}=\mu_j+\mathcal{N}\omega=\mu_{j_1}+\mathcal{N}\omega=\mu_{j_2}+\mathcal{N}\omega=\cdots=\mu_{j_s}+\mathcal{N}\omega=\lambda_{i'}+n_2\omega=\lambda_{i'_1}+n_2\omega=\lambda_{i'_2}+n_2\omega=\cdots=\lambda_{i'_{t'}}+n_2\omega$ (δ_{i1} with multiplicity-t, $\mu_j+\mathcal{N}\omega$ with multiplicity-n_1s, and $\lambda_{i'}+n_2\omega$ with multiplicity-$(\mathcal{N}-1)t'$) for $i=1,2,\ldots,n_1$, $i'=1,2,\ldots,n_1$, and $j=2,\ldots,n_2$, where $p_{i_kq}\neq 0$ and the arbitrary constants $\rho_{i_1},\ldots,\rho_{i_t},\rho_{1j_1},\ldots,\rho_{1j_s},\ldots,\rho_{n_1j_1},\ldots,\rho_{n_1j_s},\rho_{1i'_1},\ldots,\rho_{1i'_{t'}},\ldots,\rho_{(\mathcal{N}-1)i'_1},\ldots,\rho_{(\mathcal{N}-1)i'_{t'}}$ are not all zero. Thus,

$$
\begin{aligned}
&\xi^T_{\eta tst'}\mathcal{B}\\
&=\begin{bmatrix}\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\\ 1_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\\ \sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}\otimes 1_{n_2}+\sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e_l\otimes\beta_{j_h}\end{bmatrix}^T\\
&\quad\begin{bmatrix}\mathcal{B}_1 & 0 & 0\\ 0 & I_{\mathcal{N}-1}\otimes\mathcal{B}_1 & 0\\ 0 & 0 & I_{n_1}\otimes\mathcal{B}_2\end{bmatrix}\\
&=\left[\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}\right)^T\mathcal{B}_1\right.\\
&\quad\left(1_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}\otimes\alpha_{i'_{k'}}\right)^T\\
&\quad(I_{\mathcal{N}-1}\otimes\mathcal{B}_1)\\
&\quad\left.\left(\sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}\otimes 1_{n_2}+\sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e_l\otimes\beta_{j_h}\right)^T(I_{n_1}\otimes\mathcal{B}_2)\right]\\
&=\left[\left(\sum_{k=1}^{t}\rho_{i_k}\alpha^T_{i_k}+\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha^T_{i'_{k'}}\right)\mathcal{B}_1\right.\\
&\quad\left(1^T_{\mathcal{N}-1}\otimes\left(\sum_{k=1}^{t}\rho_{i_k}\alpha^T_{i_k}\mathcal{B}_1\right)-\sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e^T_{l'}\otimes\left(\alpha^T_{i'_{k'}}\mathcal{B}_1\right)\right)\\
&\quad\left.\left(\sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha^T_{i_k}\otimes\left(1^T_{n_2}\mathcal{B}_2\right)+\sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e^T_l\otimes\left(\beta^T_{j_h}\mathcal{B}_2\right)\right)\right].
\end{aligned}
\tag{12}
$$

From condition 3), we can have $\xi^T_{\eta tst'}\mathcal{B}\neq 0$, which implies that we cannot find constants $\rho_{i_1},\ldots,\rho_{i_t},\rho_{1j_1},\ldots,\rho_{1j_s},\ldots,\rho_{n_1j_1},\ldots,\rho_{n_1j_s},\rho_{1i'_1},\ldots,\rho_{1i'_{t'}},\ldots,\rho_{(\mathcal{N}-1)i'_1},\ldots,\rho_{(\mathcal{N}-1)i'_{t'}}$ (not all zero) such that $\xi^T_{\eta tst'}\mathcal{B}=0$.

Necessity: The $\mathcal{N}$-duplication weighted Corona product network (2) is controllable, and then, each left eigenvector

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is unorthogonal to $\mathcal{B}$. From (3) and (4), if $\xi_j^T\mathcal{B} \neq 0$ and $\xi_s^T\mathcal{B} \neq 0$ for $j = 2, \ldots, n_2$, it is easy to know $\beta_j^T\mathcal{B}_2 \neq 0$ and $\beta_{j_k}^T\mathcal{B}_2 \neq 0$, as well as $\mathcal{B}_2 \neq 0$. For $j = 1$, we can have that 1_{n_2} is eigenvector corresponding to eigenvalue $\mu_1 = 0$ and $1_{n_2}^T\mathcal{B}_2 \neq 0$. Therefore, $\beta_j^T\mathcal{B}_2 \neq 0$ and $\beta_{j_k}^T\mathcal{B}_2 \neq 0$ for $j = 1, 2, \ldots, n_2$, which implies that $(\mathcal{L}_2, \mathcal{B}_2)$ is controllable according to Proposition 5. Also, from (5) and (6), if $\xi_i^T\mathcal{B} \neq 0$ and $\xi_t^T\mathcal{B} \neq 0$ for $i = 1, 2, \ldots, n_1$, it is easy to know $\alpha_i^T\mathcal{B}_1 \neq 0$ and $\alpha_{i_k}^T\mathcal{B}_1 \neq 0$, which imply that $(\mathcal{L}_1, \mathcal{B}_1)$ is controllable according to Proposition 5. Finally, we consider the case if there is a common eigenvalue $\eta = \Omega_{11} \cap \Omega_2 \cap \Omega_3$. From (11), for arbitrary constants $\rho_{i_1}, \ldots, \rho_{i_t}, \rho_{1j_1}, \ldots, \rho_{1j_s}, \ldots, \rho_{n_1j_1}, \ldots, \rho_{n_1j_s}, \rho_{1i'_1}, \ldots, \rho_{1i'_{t'}}, \ldots, \rho_{(\mathcal{N}-1)i'_1}, \ldots, \rho_{(\mathcal{N}-1)i'_{t'}}$ (not all zero), there must be $\xi_{\eta ts t'}^T\mathcal{B} \neq 0$, for $i = 1, 2, \ldots, n_1, i' = 1, 2, \ldots, n_1$, and $j = 2, \ldots, n_2$, which governs that $(\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}^T + \sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}\alpha_{i'_{k'}}^T)\mathcal{B}_1 \neq 0$ or $1_{\mathcal{N}-1}^T \otimes (\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}^T\mathcal{B}_1) - \sum_{l'=1}^{\mathcal{N}-1}\sum_{k'=1}^{t'}\rho_{l'i'_{k'}}e_{l'}^T \otimes (\alpha_{i'_{k'}}^T\mathcal{B}_1) \neq 0$ or $\sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}^T \otimes (1_{n_2}^T\mathcal{B}_2) + \sum_{h=1}^{s}\sum_{l=1}^{n_1}\rho_{lj_h}e_l^T \otimes (\beta_{j_h}^T\mathcal{B}_2) \neq 0$.

Theorem 1 provides a simple and fast method for verifying the controllability and constructing controllable massive networks by simpler and lower dimensional factor networks, which greatly reduces the analyzing difficulty and computing burdens.

Theorem 2: As $\mathcal{N} = 1$, the $\mathcal{N}$-duplication weighted Corona product network (2) is controllable if and only if the following conditions hold.

1) $(\mathcal{L}_2, \mathcal{B}_2)$ is controllable.
2) If there is a common eigenvalue $\tau = \Omega_{11} \cap \Omega_2$, then $\sum_{k=1}^{t}\rho_{i_k}\alpha_{i_k}^T\mathcal{B}_1 \neq 0$ $(\exists i \in \{1, 2, \ldots, n_1\})$ or $\sum_{k=1}^{t}\rho_{i_k}p_{i_k1}\alpha_{i_k}^T \otimes (1_{n_2}^T\mathcal{B}_2) + \sum_{k=1}^{s}\sum_{i=1}^{n_1}\rho_{ij_k}e_i^T \otimes (\beta_{j_k}^T\mathcal{B}_2) \neq 0$ holds, where $\tau = \delta_{i1} = \delta_{i_t1} = \cdots = \delta_{i_t1} = \mu_{j_1} + \omega = \mu_{j_2} + \omega = \cdots = \mu_{j_s} + \omega = \mu_j + \omega$ (δ_{i1} with geometric multiplicity-t and $\mu_j + \omega$ with geometric multiplicity-$n_1 s$), and arbitrary constants $\rho_{i_1}, \ldots, \rho_{i_t}, \rho_{1j_1}, \ldots, \rho_{1j_s}, \ldots, \rho_{n_1j_1}, \ldots, \rho_{n_1j_s}$ are not all zero.

Proof: As $\mathcal{N} = 1$, according to Lemma 1, there are only two types of eigenvalues, which can be expressed as $\sigma(\mathcal{G}) = \Omega_1 \cup \Omega_2$, where $\Omega_1 = \Omega_{11} \cup \Omega_{12}$ with $\Omega_{11} = \{\delta_{i1} | \delta_{i1} = ((\omega + n_2\omega + \lambda_i) + ((\omega + n_2\omega + \lambda_i)^2 - 4\omega\lambda_i)^{1/2})/2, i = 1, 2, \ldots, n_1\}$, and $\Omega_{12} = \{\delta_{i2} | \delta_{i2} = ((\omega + n_2\omega + \lambda_i) - ((\omega + n_2\omega + \lambda_i)^2 - 4\omega\lambda_i)^{1/2})/2, i = 1, 2, \ldots, n_1\}$; and $\Omega_2 = \{\mu_j + \omega | j = 2, 3, \ldots, n_2\}$, where each eigenvalue in Ω_2 is n_1-repeated. It is easy to obtain the conclusion similar to that of Theorem 1. □

Corollary 1: As $\mathcal{N} > 1$, if $(\mathcal{L}_1, \mathcal{B}_1)$ or $(\mathcal{L}_2, \mathcal{B}_2)$ is uncontrollable, the $\mathcal{N}$-duplication weighted Corona product network (2) is uncontrollable. As $\mathcal{N} = 1$, if $(\mathcal{L}_2, \mathcal{B}_2)$ is uncontrollable, the $\mathcal{N}$-duplication weighted Corona product network (2) is uncontrollable.

Corollary 2: As $\mathcal{N} = 1$, if $(\mathcal{L}_1, \mathcal{B}_1)$ and $(\mathcal{L}_2, \mathcal{B}_2)$ are controllable, the $\mathcal{N}$-duplication weighted Corona product network (2) is controllable.

Note from Theorem 1 and Corollaries 1 and 2 that the entire composite network is controllable, and then, all the factor networks are controllable. In other words, as long as a factor network is uncontrollable, the whole composite network must be uncontrollable, which provides methods and ideas for us on how to build a controllable network model, design a controllable topology structure, and select leaders. In particular, the controllability of the entire composite network can be completely decided by the controllability of its factors from Corollary 2. Corollaries 1 and 2 can help one to test and judge the network controllability more intuitively and easily.

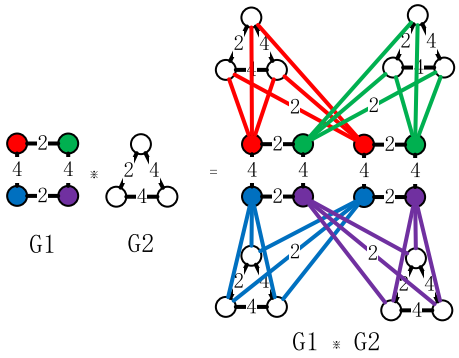


Fig. 2. 2-duplication weighted Corona product of $\mathcal{G}_1$ and $\mathcal{G}_2$.

V. Examples and Simulations

Some examples are given to verify the effectiveness of our theoretical results.

Example 1: Consider the 2-duplication weighted Corona product network of two undirected graphs as $\omega = 2$, which is shown in Fig. 2. Let

$$\mathcal{L}_1 = \begin{bmatrix} 6 & -2 & -4 & 0 \\ -2 & 6 & 0 & -4 \\ -4 & 0 & 6 & -2 \\ 0 & -4 & -2 & 6 \end{bmatrix}, \quad \mathcal{L}_2 = \begin{bmatrix} 6 & -2 & -4 \\ -2 & 6 & -4 \\ -4 & -4 & 8 \end{bmatrix}$$

$$\mathcal{B}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathcal{B}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then,

$$\mathcal{L} = \begin{bmatrix} I_2 \otimes (\mathcal{L}_1 + 6I_4) & 1_2 \otimes \left(-\left(2I_4 \otimes 1_3^T\right)\right) \\ 1_2^T \otimes (-(2I_4 \otimes 1_3)) & I_4 \otimes (\mathcal{L}_2 + 4I_3) \end{bmatrix}$$

$$\mathcal{B} = \begin{bmatrix} I_2 \otimes \mathcal{B}_1 & 0 \\ 0 & I_4 \otimes \mathcal{B}_2 \end{bmatrix}.$$

By computing, $\text{rank}(\mathcal{L}_1, \mathcal{B}_1) = 4$ and $\text{rank}(\mathcal{L}_2, \mathcal{B}_2) = 3$, and it is obvious that $(\mathcal{L}_1, \mathcal{B}_1)$ and $(\mathcal{L}_2, \mathcal{B}_2)$ are both controllable. Also, $\Omega_{11} = \{7 + (33)^{1/2}, 10, 16, 11 + (73)^{1/2}\}$, $\Omega_{12} = \{7 - (33)^{1/2}, 0, 2, 11 - (73)^{1/2}\}$, $\Omega_2 = \{4, 12, 16\}$, and $\Omega_3 = \{6, 10, 14, 18\}$, so $\Omega_{11} \cap \Omega_2 \cap \Omega_3 = \emptyset$. From Theorem 1, the 2-duplication weighted Corona product network is controllable.

From Fig. 3, we can find that all agents can reach the expected states (a rectangle configuration, denoted by "▷") from the arbitrary initial states (denoted by "∗").

Example 2: Consider the 1-duplication weighted Corona product network of two undirected graphs as $\omega = 1$, which is

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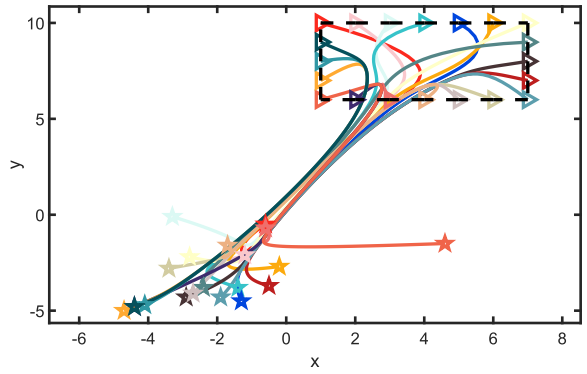


Fig. 3. Rectangle configuration for $\mathcal{G}$ in Example 1.

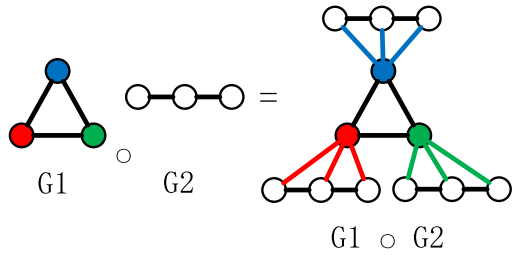


Fig. 4. 1-duplication weighted Corona product of $\mathcal{G}_1$ and $\mathcal{G}_2$.

shown in Fig. 4. Let

$$\mathcal{L}_1 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \ \mathcal{L}_2 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$
$$\mathcal{B}_1 = \mathcal{B}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then,

$$\mathcal{L} = \begin{bmatrix} \mathcal{L}_1 + 3I_3 & -\left(I_3 \otimes 1_3^T\right) \\ -(I_3 \otimes 1_3) & I_3 \otimes (\mathcal{L}_2 + I_3) \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} \mathcal{B}_1 & 0 \\ 0 & I_3 \otimes \mathcal{B}_2 \end{bmatrix}.$$

By computing, the eigenvalues of $\mathcal{L}_1$ and $\mathcal{L}_2$ are $\lambda_1 = 0, \lambda_2 = 3$, and $\lambda_3 = 3$ and $\mu_1 = 0, \mu_2 = 1$, and $\mu_3 = 3$, and the corresponding eigenvectors $\alpha_1 = [1\ 1\ 1]^T$, $\alpha_2 = [0\ -1\ 1]^T$, and $\alpha_3 = [2\ -1\ -1]^T$ and $\beta_1 = [1\ 1\ 1]^T$, $\beta_2 = [1\ 0\ -1]^T$, and $\beta_3 = [1\ -2\ 1]^T$, respectively. By taking further calculation, we can have $\Omega_{11} = \{\delta_{11} = 4, \delta_{21} = (7 + (32)^{1/2})/2, \delta_{31} = (7 + (32)^{1/2})/2\}$, $\Omega_{12} = \{\delta_{12} = 0, \delta_{22} = (7 - (32)^{1/2})/2, \delta_{32} = (7 - (32)^{1/2})/2\}$, and $\Omega_2 = \{\mu_2 + \mathcal{N}\omega = \mu_2 + 1 = 2, \mu_3 + \mathcal{N}\omega = \mu_3 + 1 = 4\}$. Obviously, $\Omega_{11} \cap \Omega_2 \neq \emptyset$ and $\alpha_1^T \mathcal{B}_1 = [1\ 1\ 0] \neq \emptyset$. Since $\text{rank}(\mathcal{L}_2, \mathcal{B}_2) = 3$, $(\mathcal{L}_2, \mathcal{B}_2)$ is controllable. From Theorem 2, this 1-duplication weighted Corona product network is controllable.

From Fig. 5, we can find that all agents can reach the expected states (a triangle configuration, denoted by "▷") from the arbitrary initial states (denoted by "∗").

Example 3: The controllability of MANs plays a fundamental and theoretical support for the control of UAV formation. Consider the formation control of a UAV Corona product network, which is shown in Fig. 6. Let

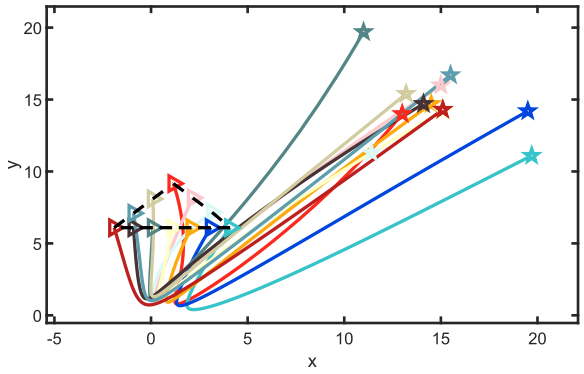


Fig. 5. Triangle configuration for $\mathcal{G}$ in Example 2.

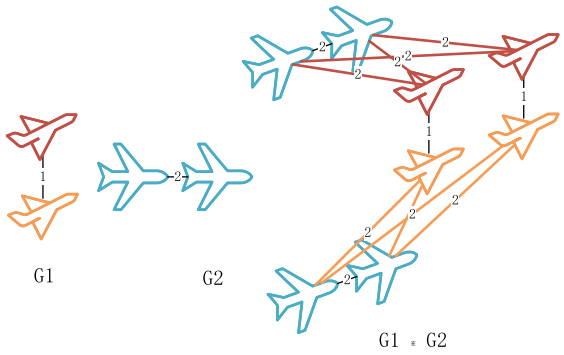


Fig. 6. UAV network topology.

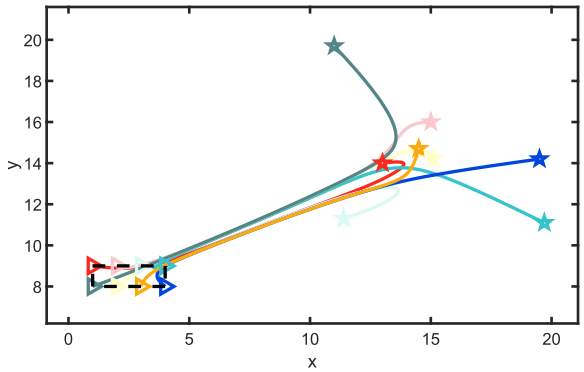


Fig. 7. Rectangle configuration for $\mathcal{G}$ in Example 3.

$$\mathcal{L}_1 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \ \mathcal{L}_2 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}; \mathcal{B}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \ \mathcal{B}_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then,

$$\mathcal{L} = \begin{bmatrix} \mathcal{L}_1 + 2I_2 & -\left(I_2 \otimes 1_2^T\right) \\ -(I_2 \otimes 1_2) & I_2 \otimes (\mathcal{L}_2 + I_2) \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} I_2 \otimes \mathcal{B}_1 & 0 \\ 0 & I_2 \otimes \mathcal{B}_2 \end{bmatrix}.$$

It is easy to see that $(\mathcal{L}_1, \mathcal{B}_1)$ and $(\mathcal{L}_2, \mathcal{B}_2)$ are controllable; therefore, the UAV network is controllable.

From Fig. 7, we can find that all agents can reach the expected states (a rectangle configuration, denoted by "▷") from the arbitrary initial states (denoted by "∗").

One important thing to note is that, theoretically, the obtained results of this work are still valid if the network adds some agents (vertices) and edge connections, as long as the

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controllability conditions are satisfied. However, in the actual simulation process, even adding an agent or a connection edge will bring great difficulties to design the controllable network topology structure and select the leader agents. Moreover, from Examples 1 to 3, due to the construction of $\mathcal{N}$-duplication weighted Corona product networks, even though the number of agents in the factor networks is small and agents' interactions are relatively simple, the number of agents in the composite network will increase on a large scale and the interaction topology will be very complex, both of which will improve the difficulty of simulation.

VI. Conclusion

This work has constructed a new composite network via Corona product, analyzed the controllability of the network, and derived some necessary and (or) sufficient conditions on the controllability, which are valid and easy to compute and verify. This work considers the case that the interaction topology is undirected and the weights between different factor networks are identical and positive, which will be extended to the directed graph and general weights in our future work. In addition, another future work will consider a more relevant and complex case, that is, how to select the leaders or external inputs to make Corona product MANs controllable.

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Bo Liu received the Ph.D. degree in dynamics and control from Peking University, Beijing, China, in 2007.

She is currently a Professor with the Minzu University of China, Beijing. Her research interests include swarm dynamics, networked systems, collective behavior, and coordinate control of multiagent systems.



Xuan Li was born in 1999. He received the B.S. degree in computer science and technology from the Wuhan University of Science and Technology, Wuhan, China, in 2021. He is currently pursuing the M.S. degree in computer science and technology at the Minzu University of China, Beijing, China.

His research interests are in the fields of control of multiagent systems and network controllability.



Junjie Huang received the B.S. degree in mathematics from Inner Mongolia Normal University, Hohhot, China, in 2000, and the M.S. and Ph.D. degrees in applied mathematics from Inner Mongolia University, Hohhot, in 2002 and 2006, respectively.

Since 2008, he has been a Full Professor with the School of Mathematical Sciences, Inner Mongolia University. His research interests lie in the areas of operator theory and its applications in mathematical physics, mechanics, and engineering.



Housheng Su received the B.S. degree in automatic control and the M.S. degree in control theory and control engineering from the Wuhan University of Technology, Wuhan, China, in 2002 and 2005, respectively, and the Ph.D. degree in control theory and control engineering from Shanghai Jiao Tong University, Shanghai, China, in 2008.

From December 2008 to January 2010, he was a Post-Doctoral Researcher with the Department of Electronic Engineering, City University of Hong Kong, Hong Kong. Since November 2014, he has been a Full Professor with the School of Automation, Huazhong University of Science and Technology, Wuhan. His research interests lie in the areas of multiagent coordination control theory and its applications to autonomous robotics and mobile sensor networks.

Dr. Su is an Associate Editor of *IET Control Theory and Applications*.

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